

SUBSYSTEM REQUIREMENTS SPECIFICATION INTEGRATED PLATFORM SERVICES (IPS)

Digital Interlocking OBB

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	305	16282~0241	01 RL
3BU 17300 2051 DTAPA			

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LOG OF CHANGES			
Edition	Date	Author	Change Description
01	2025-12-04	Karmatska N.	First edition. Details see 4.1

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1. INTRODUCTION

1.1 PURPOSE

The present document contains the requirements of the subsystem “Integrated Platform Services” (IPS) for Digital Interlocking OBB.

1.2 INTENDED AUDIENCE

- Product Design Authority (PDA)
- Project Manager (PM)
- Solution Engineering Manager (SEM)
- Project Security Manager (PSM)
- Product Line Manager (PLM)
- Test Manager (TM)
- Validator (VAL)
- Verifier (VER)
- Safety Assurance Manager (SAM)
- Quality Assurance Manager (QAM)
- RAM Assurance Manager (RAM-AM)
- System Architect (ARC)
- Work Package Manager (WPM)

1.3 DISTRIBUTION

A new version of the document is distributed to all persons according to chapter 1.2

1.4 DOCUMENT MANAGEMENT

See Document Management Rules [DMR] for a description of the SSS document management.

1.5 REQUIREMENTS MANAGEMENT

See Requirements Management Plans [IPS_RMP] and [DIL_RMP] for a description of the SSS Requirements management.

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1.6 REFERENCED DOCUMENTS

1.6.1 PLANNING DOCUMENTS

[EL_GLOSS]	Glossary: Definitions, Acronyms and Abbreviations - ELEKTRA 2.x System 3BU 15000 0081 DCAPA, Ed.06
[DIL_RMP]	Requirements Management Plan – Digital Interlocking OBB, 3BU 17300 1207 DUAPA, Ed.01
[IPS_RMP]	Requirements Management Plan – IPS Solution Element, 3BU 17300 2010 DUAPA, Ed.01
[DMR]	Document Management Rules 83530061-DOC-AUT-013
[G-COSM]	Guide for software cots and open source management 61565035-DDQ-GRP-005

Table 1: Planning documents

1.6.2 STANDARDS

[EN-50716]	European Standard “Railway applications – Requirements for software development”, EN 50716:2023
[EN-50129-2018]	European Standard “Railway applications – Communication, signalling and processing systems – Safety related electronic systems for signaling”, EN 50129:2018 + AC:2019
[EN-50129-2010]	EN 50159:2010 +A1:2020 – Railway application – Communication, signalling and processing: Safety-related communication in transmission systems
[EN 50128]	EN 50128:2011 +A1:2020 +A2:2020 – Railway application – Communication, signalling and processing systems – Software for railway control and protection systems
[IEC 62443-2-4]	IEC 62443-2-4:2015/AMD 1:2017 Security for industrial automation and control systems – Part 2-4: Security program requirements for IACS service providers
[IEC 62443-4-1]	IEC 62443-4-1:2018-01 Security for industrial automation and control systems –Part 4-1: Secure product development lifecycle requirements
[IEC 62443-4-2]	IEC 62443-4-2:2020-01 Security for industrial automation and control systems Part 4-2: Technical security requirements for IACS components
[EU-R SPCS]	Europe’s Rail System Pillar Cybersecurity Specification
[RFC 5424]	The Syslog Protocol: https://datatracker.ietf.org/doc/html/rfc5424
[RFC 5425]	Transport Layer Security (TLS) Transport Mapping for Syslog: https://datatracker.ietf.org/doc/html/rfc5425

Table 2: Standards

1.6.3 REQUIREMENTS AND TECHNICAL REFERENCES

In the following table, some references to documents originally created for OMC 3.0 Product are provided. These documents, such as [25], [27] or [29], contain information that may be partly or completely adapted for the DIL product. Therefore, the references to the OMC 3.0 documentation will either be updated to include DIL related content or be entirely replaced with a new document created specifically for DIL.

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[01]	System Requirements Specification - Digital Interlocking OBB (DIL) - General Requirements 3BU 17300 1251 DTAPA, Ed.01 System Requirements Specification - Digital Interlocking OBB (DIL) - Central Unit Part 1 3BU 17300 1255 DTAPA Ed.01 System Requirements Specification - Digital Interlocking OBB (DIL) - Field Element Controller and Stations 3BU 17300 1256 DTAPA Ed.01 System Requirements Specification - Digital Interlocking OBB (DIL) – RAMS 3BU 17300 1257 DTAPA Ed.01
[02]	Interface Control Description, IPS Cloud, IF-70 Retrofit (RETROFIT Scripts - IPS Cloud) 3BU 51799 0003 PBAPA, Ed.01(planned)
[03]	Interface Control Description, IPS Cloud, IF-71 Container Delivery Definition 3BU 51799 0004 PBAPA, Ed.01 (planned)
[04]	Interface Design Description, ADM-Prometheus 3BU 51799 0005 PBAPA, Ed.01 (planned)
[05]	< remains intentionally blank >
[06]	System Subsystem Specification, Generic Server and Network Cabinet 3BU 16000 0002 DTAPA, Ed.04
[07]	Software Design Description, OMC external interfaces 3BU 51500 1238 PEAPA, Ed.03
[08]	Detailed Design Document, DIL-IPS Low Level Design (Network LLD) 3BU 17300 2054 EDAPA, Ed.01 (planned)
[09]	Interface Control Description, IPS Cloud, IF-73 Grafana (ADM - IPS Cloud) 3BU 51799 0006 PBAPA, Ed.01 (planned)
[10]	System Segment Specification (SSS) - DIL-IPS Workflows 3BU 17300 2055 DTAPA, Ed.01 (planned)
[11]	< remains intentionally blank >
[12]	System Subsystem Design Description, Generic Server and Network Cabinet 3BU 16000 0003 DSAPA, Ed.03
[13]	< remains intentionally blank >
[14]	Digitales Stellwerk - Technische Konzepte 3BU 17300 0010 DSAPC, Ed.04
[15]	System Segment Specification, IPS OMC 3BU 51500 3200 DTAPA, Ed.03
[16]	Feature List DSTW 3BU 17300 0000 FLAPC Ed.01 Pd.01
[17]	< remains intentionally blank >
[18]	< remains intentionally blank >
[19]	< remains intentionally blank >
[20]	< remains intentionally blank >
[21]	< remains intentionally blank >
[22]	< remains intentionally blank >
[23]	< remains intentionally blank >
[24]	System Subsystem Design Description, Core Network Design Description 3BU 17300 2056 DSAPA, Ed.01 (planned)
[25]	System Design Description, Monitoring Concept OMC 3.x 3BU 51500 2048 PEAPA, Ed.03
[26]	Evidence of Compliance, Evidence against incoming SRACs, 3BU 17300 xxxx DUAPA, Ed.01 (planned)
[27]	Software Design Description, Logging Concept 3BU 51500 2061 PEAPA, Ed.01

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[28]	System Subsystem Design Description, IPS 3BU 51500 3201 DSAPA, Ed.03 Pd.01
[29]	System Design Description, Users and Roles Management OMC 3.x 3BU 51500 2052 PEAPA, Ed.03
[30]	OS Hardening Rules, CIS Ubuntu Linux Benchmark 3BU 51799 0008 REAPA, Ed.01 (planned)
[31]	User Manual, Nokia NSP User Documentation 3BU 51799 0009 PCAPA, Ed.01 (planned)
[32]	Installation Guide IPS 3BU 51799 0010 RJAPA, Ed.01 (planned)
[33]	System Subsystem Design Description (SSDD) - DIL-IPS Workflows 3BU 17300 2057 PEAPA, Ed.01 (planned)
[34]	Software Design Description - DIL-IPS Workflows 3BU 17300 2058 PEAPA, Ed.01 (planned)
[35]	< remains intentionally blank >
[36]	< remains intentionally blank >
[37]	Interface Design Description, OMC 2.0, IF-25 Monitoring Stromversorgung 3BU 51500 1140 PBAPC, Ed.02
[38]	< remains intentionally blank >
[39]	< remains intentionally blank >
[40]	Rollout Concept 3BU 51500 2058 EDAPA Ed.01
[41]	< remains intentionally blank >
[42]	< remains intentionally blank >
[43]	SSRS NMR - TAS Platform Core SW - 2.x 3CR 00055 7161 DTAPA, Ed.08

Table 3: Requirements and Technical References

1.7 ABBREVIATIONS, ACRONYMS, TERMS AND DEFINITIONS

See Definitions, Acronyms and Abbreviations [EL_GLOSS].

Additional abbreviations not mentioned there: see table below.

API	Application Programming Interface
CA	Certificate Authority
CDS	Cluster Deployment Supervisor
CE	Computing Element
CG	Computing Group
CN	Computing Node
COTS	Commercial off-the-shelf
CSR	Certificate Signing Request
DIL	Digital Interlocking OBB (System Level)
EP	ELEKTRA Personality (designed for non-redundant operation)
EP-red	EP redundant
FRACAS	Failure Reporting, Analysis, and Corrective Action System
HMD	Health Monitor Daemon
GDP	GTS Digital Platform
IIS	Infrastructure Information Service
IPS	Integrated Platform Services

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K3s	K3s is a highly available, certified and lightweight Kubernetes distribution. See https://k3s.io/ .
K8s	Also known as Kubernetes. Therefore, refer to Kubernetes.
KPI	Key Performance Indicator
Kubernetes	Kubernetes, abbreviated as K8s, is an open source system for automating deployment, scaling, and management of containerized applications across multiple hosts. See https://kubernetes.io/ .
LSP	Label Switched Path
ML	Maturity Level
MTU	Maximum Transmission Unit (defines the maximum amount of data that can be sent in a single frame or packet at the Data Link layer (Layer 2) before being divided into smaller parts if necessary)
MPLS	Multiprotocol Label Switching
NMR	N-Modular Redundancy. A software component of TAS Platform Core. Implemented as 2 or 4 times 2-out-of-2.
NSP	Network Services Platform™ tool from NOKIA
PHC	Physical Hardware Clock
PoS	Point of Signaling
RSVP	Resource Reservation Protocol
SL	Security Level
SIL	Safety Integrity Level
SRAC	Safety-related Application Condition
TCP	Transmission Control Protocol
TLS	Transport Layer Security
VIL	Virtual Interlocking (Subsystem Level)
VM	Virtual Machine
VPLS	Virtual Private LAN Services

Table 4: Abbreviations and Acronyms

Terms and Definitions are provided below (non-exhaustive list)

Application workloads	Refers to the set of DIL applications, processes or tasks that a system or IPS platform needs to handle. These workloads can include software applications, databases, or services that are running and consuming system resources (CPU, memory, storage) on the platform
Air-gapped environment	Refers to a network or system that is isolated from untrusted networks including the internet, reducing the possibility that sensitive data or critical applications are exposed to external threats such as cyberattacks.
An isolated network	A network that is logically or physically separated from other networks to prevent unauthorized access and protect data, with limited or controlled external connections.
Balanced distribution	A distribution of k items to n buckets is called balanced if the number of items per bucket differs by at most 1.
Balanced increment	An increase in a balanced distribution by one item, so that the resulting distribution remains balanced.
Building block	A building block is a functional cluster grouping related components and services that together deliver a specific capability within the IPS Subsystem architecture.

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Interconnection LAN	Geo-redundant network infrastructure established between geographically separated sites of the System.
maintenance workflows	Refers to a structured process or series of steps that are followed to maintain and support a system, applications, or infrastructure
MPLS	Multiprotocol Label Switching (MPLS) is a versatile networking technology designed to improve packet forwarding efficiency in modern networks. It achieves this by introducing labels that are attached to packets, allowing routers to make forwarding decisions based on these labels rather than traditional IP header information. MPLS introduces labels that routers use to forward packets, allowing for faster and more efficient routing decisions.
RSVP	Resource Reservation Protocol (RSVP) is a signaling protocol used in conjunction with MPLS to enable dynamic resource allocation along a network path. Quality of Service (QoS) allows applications and users to request specific qualities of service, ensuring reliable and predictable data transmission
VPLS	Virtual Private LAN Services is a service that emulates a Local Area Network (LAN) over a Wide Area Network (WAN)m enabling geographically dispersed locations to function as a single Virtual LAN
LSP	Label Switched Path (LSP) is a predetermined path through the network, established using MPLS, to ensure efficient data forwarding. LSPs enable end-to-end communication with guaranteed QoS, making them particularly useful for critical applications and services.
Network monitoring	Refers to continuous monitoring of the network status to identify and detect problems
Network diagnosis	Refers to detailed data analysis to find the cause and eliminate network problems
non-safety-related	Non-safety related system is a system whose functions have no direct impact on safety of railway operations
Upgrade	An upgrade includes the following: Introducing a new baseline to a VIL for the purpose of bug fixes, expanding functionality, or both. Introducing new configuration data (station data). Installing new firmware on the Object Controller or any combination of the above.
Patch	Changes or extensions to functions that are not covered by the "upgrade" list and can be performed without interruption. Example: Introducing a new security patch or changing an application property (changes to settings such as memory size, buffer sizes, timeouts, etc.)
PoS	A network interface or component responsible for the exchange of signaling messages between subsystem within the railway signalling system.
1x redundancy	A system design where a single backup component or path is provided to ensure reliability in case of failure. A single level of redundancy for example for a network connection means that for every primary connection there is one backup connection
2x redundancy	A system design where two backup components or paths are included enhancing fault tolerance and increasing system reliability. 2x redundant network means that for every primary connection there are two backup connections.

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Two-fold redundancy Technical equivalent of 2x redundancy

Geo-redundancy A system design approach that distributes critical components, data or infrastructure across multiple geographic locations (Datacenters) to ensure availability, fault tolerance, and disaster recovery in case of one location (Datacenter) failure

1.8 STRUCTURE OF THE REQUIREMENTS

See Plans [IPS_RMP] and [DIL_RMP] for a description of the foreseen SSS requirement structure.

1.9 INFORMATION FOR DOCUMENT GENERATION

The present document was generated by using IBM™ Rational DOORS.

Settings applied:

DOORS/RMF Version:	/RMF
DOORS/RMF Module Name:	EL4 IPS SSS
DOORS/RMF Module Version (Baseline):	1.0
DOORS/RMF Module Path:	/ELEKTRA 4.0/Specification
DOORS/RMF Export View:	WEXP SSS (IPS-DIL) 2051
DOORS/RMF Current Filter:	(TA Req Status != Deleted) AND (TA SRS includes 2051)
DOORS/RMF Export Options 1:	2,false,0,0,false,false,0,true,false,
DOORS/RMF Export Options 2:	true,false,true,2,false,false,true,PostProcessAllStyles,\\Thales-AT-Data\Thales-AT\PROJECTS\ReqMng\DOORS\WordExport\DOORS-EXPORT-TEMPLATES\EL4\SRS\Exports\WEXP_6801.log,
DOORS/RMF Export Options 3:	\\Thales-AT-Data\Thales-AT\PROJECTS\ReqMng\DOORS\WordExport\DOORS-EXPORT-TEMPLATES\EL4\SRS\Exports\WEXP_6801.log,true,true,true,false,false,true,0,true,4,
DOORS/RMF Export Options 4:	0,###,===,true,false,false,true,true,
Documentation-Vob Template Name:	3BU_17300_2051_DTAPA-Template

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2. OVERVIEW

2.1 INTRODUCTION

The goal of IPS is to provide infrastructure for Hitachi Rail GTS Austria GmbH products in a uniform way. Digital Interlocking uses the functionality provided by IPS. IPS provides infrastructure as a service (IaaS), a platform that provides computing, networking and storage infrastructure. In addition, IPS provides a platform as a service (PaaS) where container as a service (CaaS) is one of the services to be used by Digital Interlocking.

The following figure displays a conceptual overview of IPS used by the DIL system.

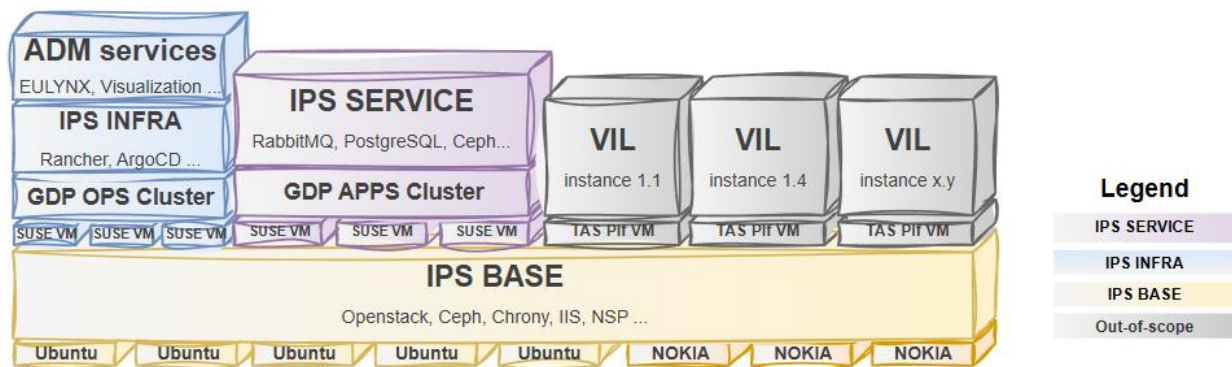


Figure 1: System overview per DIL location

The different boxes represent functional building blocks:

IPS BASE (shown in yellow) provides infrastructure as a service (IaaS) comprising networking, computing and storage.

IPS INFRA (shown in blue) provides services to monitor, manage and administrate IPS SERVICE clusters and the IPS INFRA building block itself:

- GDP services is a GTS Digital Platform solution intended to provide a set of components serving as building blocks for creating modern, robust, secure & scalable applications inside Hitachi Rail GTS Austria
- IPS services (shown in blue).

IPS SERVICE (shown in violet) consists of GDP services to run containerized applications specific to the DIL system.

IPS NETWORK MONITORING (not illustrated) provides services to monitor, manage and administrate the whole network infrastructure (Core and Interconnection networks). This building block uses the [NSP] tool which is a distributed system that provides comprehensive infrastructure, service management, and resource control within and among the whole network infrastructure (Core and interconnection networks).

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3. REQUIREMENTS

The Legacy ID attribute indicates that the requirement was inherited from the [OMC] project and will be reused in DIL.

3.1 GENERAL FUNCTIONAL REQUIREMENTS

This chapter lists requirements that relate to more than one IPS building block.

EL4-IPS-SSS-REQ-00001- MCR Support COTS server & networking hardware			
[IPS] shall support standard [COTS] server and network hardware and their components without requiring any proprietary configuration or modification.			
Comments: For more details refer to [G-COSM].			
Legacy ID:		Priority:	Mandatory
Status:	Approved	Origin:	EL4-DIL-SSS-REQ-00448 EL4-DIL-SSS-REQ-00481 EL4-DIL-SSS-REQ-00355 EL4-DIL-SSS-REQ-00356
Target Release:	IPSS OBB 1.0.0	Scope:	Sub-System
See also:	[14] Chapter 6.4.4 [14] Chapter 13.4.2 [14] Chapter 16.1.3	Originator:	Technics (R&D)
Verification method:	Test	Verification step:	Integration test
Issues/Decisions:			

EL4-IPS-SSS-REQ-00003- PER [IPS] services an isolated network			
[IPS] shall support installation, patching and upgrading of services in an isolated network			
Comments:			
Legacy ID:		Priority:	Mandatory
Status:	Approved	Origin:	
Target Release:	IPSS OBB 1.0.0	Scope:	Sub-System
See also:		Originator:	Technics (R&D)
Verification method:	Test	Verification step:	System test
Issues/Decisions:			

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EL4-IPS-SSS-REQ-00004- PER Applications workloads an isolated network	
[IPS] shall provide services for application workloads to be installed, upgraded, patched and run in an isolated network	
<i>Comments:</i>	
<i>Legacy ID:</i>	<i>Priority:</i> Mandatory
<i>Status:</i> Approved	<i>Origin:</i>
<i>Target Release:</i> IPSS OBB 1.0.0	<i>Scope:</i> Sub-System
<i>See also:</i>	<i>Originator:</i> Technics (R&D)
<i>Verification method:</i> Test	<i>Verification step:</i> System test
<i>Issues/Decisions:</i>	

EL4-IPS-SSS-REQ-00005- PSR IF-22 Printer Interface disabling	
[IPS] shall provide the capability to enable or disable the printing functionality available through IF-22	
<i>Comments:</i> IF-22 is defined in the System Segment Specification, IPS OMC [15].	
<i>Legacy ID:</i>	<i>Priority:</i> Mandatory
<i>Status:</i> Approved	<i>Origin:</i> EL4-FL-FEA-0015
<i>Target Release:</i> IPSS OBB 1.0.0	<i>Scope:</i> Sub-System
<i>See also:</i>	<i>Originator:</i> Technics (R&D)
<i>Verification method:</i> Test	<i>Verification step:</i> Integration test
<i>Issues/Decisions:</i>	

EL4-IPS-SSS-REQ-00010- DPR Centralized alerting capabilities	
[IPS] shall provide redundant centralized alerting capabilities for infrastructure, networks and applications	
<i>Comments:</i>	
<i>Legacy ID:</i>	<i>Priority:</i> Mandatory
<i>Status:</i> In negotiation	<i>Origin:</i> EL4-DIL-SSS-REQ-00319 EL4-DIL-SSS-REQ-00321 EL4-DIL-SSS-REQ-00990
<i>Target Release:</i> IPSS OBB 1.0.0	<i>Scope:</i> Sub-System
<i>See also:</i>	<i>Originator:</i> Technics (R&D)
<i>Verification method:</i> Test	<i>Verification step:</i> Integration test
<i>Issues/Decisions:</i>	

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EL4-IPS-SSS-REQ-00012- Configuration of the backup service DAT	
The [IPS] backup service shall be configurable to support the following backup capabilities: •frequency: it shall be possible to set backup intervals •retention: it shall be possible to define retention periods for backup •data: it shall be possible to specify which resources are to be backed up, including file system, blob and DB storage •backup location: [IPS] shall allow backups to be stored in the local servers or geo-distributed data centers	
Comments:	
Legacy ID:	Priority: Mandatory
Status: Approved	Origin:
Target Release: IPSS OBB 1.0.0	Scope: Sub-System
See also:	Originator: Technics (R&D)
Verification method: Test	Verification step: Integration test
Issues/Decisions:	

EL4-IPS-SSS-REQ-00013- [IPS] georedundant setup DPR	
[IPS] shall provide highly available geo-redundant setup for all infrastructure and services including compute, storage and network, across up to 3 geographically separated DC locations.	
Comments:	
Legacy ID:	Priority: Mandatory
Status: Approved	Origin: EL4-DIL-SSS-REQ-00456
Target Release: IPSS OBB 2.0.0	Scope: Sub-System
See also:	Originator: Technics (R&D)
Verification method: Test	Verification step: System test
Issues/Decisions:	

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EL4-IPS-SSS-REQ-00014- DPR Configuration for virtual Network	
[IPS] shall provide its own [SMI], [SDI] and [SSI] traffic with customer-defined, configurable DSCP markers.	
<i>Comments:</i>	
<i>Legacy ID:</i> <i>Status:</i> Approved <i>Target Release:</i> IPSS OBB 1.0.0 <i>See also:</i> EL4-IPS-SSS-REQ-90037 <i>Verification method:</i> Test	<i>Priority:</i> Mandatory <i>Origin:</i> EL4-DIL-SSS-REQ-00544 EL4-DIL-SSS-REQ-00545 EL4-DIL-SSS-REQ-00548 <i>Scope:</i> Sub-System <i>Originator:</i> Technics (R&D) <i>Verification step:</i> Integration test
<i>Issues/Decisions:</i>	

EL4-IPS-SSS-REQ-00017- FUR [DSCP] traffic markers	
The traffic on the [SCI], [SMI], [SDI] and [SSI] interfaces shall be provided with a [DSCP] traffic marker, which are configurable using network technologies specified in the [EL4-IPS-SSS-REQ-04021].	
<i>Comments:</i> [EU-R SPCS] requires VLAN separation, but the customer explicitly specifies [EU-R SPCS] deviation. See LH 04_709 and D&JF 73	
<i>Legacy ID:</i> <i>Status:</i> Approved <i>Target Release:</i> IPSS OBB 1.0.0 <i>See also:</i> [14] Chapter 1.3.4 <i>Verification method:</i> Test	<i>Priority:</i> Mandatory <i>Origin:</i> EL4-DIL-SSS-REQ-00548 EL4-DIL-SSS-REQ-00545 <i>Scope:</i> Sub-System <i>Originator:</i> Technics (R&D) <i>Verification step:</i> System test
<i>Issues/Decisions:</i>	

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EL4-IPS-SSS-REQ-00018- [DSCP] markers for network prioritization			
FUR			
[IPS] shall ensure that prioritization of the network traffic on interfaces using [DSCP] markers is compliant with [Eu.Pos.1140]			
<i>Comments:</i>			
<i>Legacy ID:</i>		<i>Priority:</i>	Mandatory
<i>Status:</i>	Approved	<i>Origin:</i>	EL4-DIL-SSS-REQ-00544 EL4-DIL-SSS-REQ-00545 EL4-DIL-SSS-REQ-00548
<i>Target Release:</i>	IPSS OBB 1.0.0	<i>Scope:</i>	Sub-System
<i>See also:</i>		<i>Originator:</i>	Technics (R&D)
<i>Verification method:</i>	Test	<i>Verification step:</i>	System test
<i>Issues/Decisions:</i>			

EL4-IPS-SSS-REQ-00021- Automatic switch-over of Geo-redundant connections			
DPR			
[IPS] shall provide automatic switch-over to geo-redundant connections in the event of primary connection failure.			
<i>Comments:</i>			
<i>Legacy ID:</i>		<i>Priority:</i>	Mandatory
<i>Status:</i>	In negotiation	<i>Origin:</i>	
<i>Target Release:</i>	IPSS OBB 2.0.0	<i>Scope:</i>	Sub-System
<i>See also:</i>	[14] Chapter 1.4.5	<i>Originator:</i>	Technics (R&D)
<i>Verification method:</i>	Test	<i>Verification step:</i>	System test
<i>Issues/Decisions:</i>			

EL4-IPS-SSS-REQ-00022- Manual switch-over of Geo-redundant connections			
DPR			
[IPS] shall provide manual switch-over capabilities to geo-redundant location, allowing to re-route network traffic in the event of primary connection failure.			
<i>Comments:</i> This functionality is available only for users with the Role = Admin or other roles granted with manual switch-over permissions as defined in the RBAC configuration.			
<i>Legacy ID:</i>		<i>Priority:</i>	Mandatory
<i>Status:</i>	In negotiation	<i>Origin:</i>	EL4-DIL-SSS-REQ-00700
<i>Target Release:</i>	IPSS OBB 2.0.0	<i>Scope:</i>	Sub-System
<i>See also:</i>	[14] Chapter 1.4.5	<i>Originator:</i>	Technics (R&D)
<i>Verification method:</i>	Test	<i>Verification step:</i>	System test
<i>Issues/Decisions:</i>			

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EL4-IPS-SSS-REQ-00024- Geo-redundant storage DPR	
[IPS] shall support geo-redundant, consistent configuration and application binary storage. Configurations, VMs and Images shall be synchronized between datacenters.	
<i>Comments:</i>	
<i>Legacy ID:</i> <i>Status:</i> Approved <i>Target Release:</i> IPSS OBB 2.0.0 <i>See also:</i> <i>Verification method:</i> Test	<i>Priority:</i> Mandatory <i>Origin:</i> EL4-DIL-SSS-REQ-00456 EL4-DIL-SSS-REQ-00457 EL4-DIL-SSS-REQ-00352 <i>Scope:</i> Sub-System <i>Originator:</i> Technics (R&D) <i>Verification step:</i> System test
<i>Issues/Decisions:</i>	

EL4-IPS-SSS-REQ-00025- Local storage DPR	
[IPS] shall support local consistent configuration and application binary storage. Configurations, VMs and Images shall be available.	
<i>Comments:</i>	
<i>Legacy ID:</i> <i>Status:</i> Approved <i>Target Release:</i> IPSS OBB 1.0.0 <i>See also:</i> <i>Verification method:</i> Test	<i>Priority:</i> Mandatory <i>Origin:</i> EL4-IDJ-RA:ID:0005 EL4-DIL-SSS-REQ-00352 <i>Scope:</i> Sub-System <i>Originator:</i> Technics (R&D) <i>Verification step:</i> System test
<i>Issues/Decisions:</i> EL4-IDJ-RA:ID:0005 Consistent data within and across [Datacenter]s Decision: IPS needs to guarantee that data is consistent across all Availability Zones and datacenters.	

EL4-IPS-SSS-REQ-00029- Monitoring, diagnosis and logging data MDR	
[IPS] shall provide monitoring and diagnosis of log data.	
<i>Comments:</i> Considering [RFC 5424]; [RFC 5425]	
<i>Legacy ID:</i> <i>Status:</i> In negotiation <i>Target Release:</i> IPSS OBB 1.0.0 <i>See also:</i> <i>Verification method:</i> Test	<i>Priority:</i> Mandatory <i>Origin:</i> EL4-DIL-SSS-REQ-00476 <i>Scope:</i> Sub-System <i>Originator:</i> Technics (R&D) <i>Verification step:</i> Integration test
<i>Issues/Decisions:</i>	

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EL4-IPS-SSS-REQ-00044- MDR Data transmit via syslog	
[IPS] shall collect and transmit data using syslog.	
Comments: Considering [RFC 5424]; [RFC 5425]	
Legacy ID:	Priority: Mandatory
Status: In negotiation	Origin: EL4-DIL-SSS-REQ-00476
Target Release: IPSS OBB 1.0.0	Scope: Sub-System
See also: EL4-IPS-SSS-REQ-90077	Originator: Technics (R&D)
Verification method: Test	Verification step: Integration test
Issues/Decisions:	

EL4-IPS-SSS-REQ-00031- MDR Monitoring and status information of hardware, server and application workloads	
[IPS] shall collect monitoring and status information of hardware and application workloads.	
Comments: [ADM] receives the information from [IPS]	
Legacy ID:	Priority: Mandatory
Status: Approved	Origin: EL4-DIL-SSS-REQ-00321 EL4-DIL-SSS-REQ-00990 EL4-DIL-SSS-REQ-00358
Target Release: IPSS OBB 1.0.0	Scope: Sub-System
See also:	Originator: Technics (R&D)
Verification method: Test	Verification step: System test
Issues/Decisions:	

EL4-IPS-SSS-REQ-00047- FUR Environmental Conditions - Monitoring of Warning and Limit Values	
[IPS] shall monitor the environmental conditions according to chapter "[Environmental Conditions]" with regard to warning and limit values.	
Comments:	
Legacy ID:	Priority: Mandatory
Status: Approved	Origin: EL4-DIL-SSS-REQ-00758 EL4-DIL-SSS-REQ-00759 EL4-DIL-SSS-REQ-00811
Target Release: IPSS OBB 1.0.0	Scope: Sub-System
See also:	Originator: Technics (R&D)
Verification method: Test	Verification step: System test
Issues/Decisions:	

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EL4-IPS-SSS-REQ-00032- MDR Alert on monitoring and status information triggers			
[IPS] shall alert on defined monitoring and status information triggers.			
<i>Comments:</i> For more details refer to IF-72 [04].			
<i>Legacy ID:</i>		<i>Priority:</i>	Mandatory
<i>Status:</i>	In negotiation	<i>Origin:</i>	EL4-DIL-SSS-REQ-00321 EL4-DIL-SSS-REQ-00990 EL4-DIL-SSS-REQ-00759 EL4-DIL-SSS-REQ-00811 EL4-DIL-SSS-REQ-00358
<i>Target Release:</i>	IPSS OBB 1.0.0	<i>Scope:</i>	Sub-System
<i>See also:</i>		<i>Originator:</i>	Technics (R&D)
<i>Verification method:</i>	Test	<i>Verification step:</i>	Integration test
<i>Issues/Decisions:</i>			

EL4-IPS-SSS-REQ-00033- PRR Performance and capacity requirements			
[IPS] shall support performance and capacity requirements defined by the projects / applications.			
<i>Comments:</i>			
<i>Legacy ID:</i>		<i>Priority:</i>	Mandatory
<i>Status:</i>	In negotiation	<i>Origin:</i>	EL4-FL-FEA-0015
<i>Target Release:</i>	IPSS OBB 1.0.0	<i>Scope:</i>	Sub-System
<i>See also:</i>		<i>Originator:</i>	Technics (R&D)
<i>Verification method:</i>	Test	<i>Verification step:</i>	Integration test
<i>Issues/Decisions:</i>			

EL4-IPS-SSS-REQ-00034- MDR Mechanisms for power outages of IPS - Services			
[IPS] shall provide shut-down and start-up mechanisms for [IPS] services after entire site [Datacenter] power outages.			
<i>Comments:</i>			
<i>Legacy ID:</i>		<i>Priority:</i>	Mandatory
<i>Status:</i>	Approved	<i>Origin:</i>	EL4-DIL-SSS-REQ-00655
<i>Target Release:</i>	IPSS OBB 1.0.0	<i>Scope:</i>	Sub-System
<i>See also:</i>		<i>Originator:</i>	Technics (R&D)
<i>Verification method:</i>	Test	<i>Verification step:</i>	System test
<i>Issues/Decisions:</i>			

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EL4-IPS-SSS-REQ-00043-MDR Mechanisms for power outages of IPS - [VM]s and containers	
[IPS] shall provide shut-down and start-up mechanisms for [VM]s and application containers running on top of [IPS] after entire site [Datacenter] power outages.	
<i>Comments:</i>	
<i>Legacy ID:</i>	<i>Priority:</i> Mandatory
<i>Status:</i> Approved	<i>Origin:</i> EL4-DIL-SSS-REQ-00655
<i>Target Release:</i> IPSS OBB 1.0.0	<i>Scope:</i> Sub-System
<i>See also:</i>	<i>Originator:</i> Technics (R&D)
<i>Verification method:</i> Test	<i>Verification step:</i> System test
<i>Issues/Decisions:</i>	

EL4-IPS-SSS-REQ-04005-FUR Virtual Machines for containerized non-safety-related applications on Clusters	
OpenStack projects shall manage [VM]s to run containerized non-safety-related applications.	
<i>Comments:</i>	
<i>Legacy ID:</i> IPS:SSS:REQ:4005	<i>Priority:</i> Mandatory
<i>Status:</i> Approved	<i>Origin:</i> EL4-FL-FEA-0015
<i>Target Release:</i> IPSS OBB 1.0.0 BFZ-3.0	<i>Scope:</i> Sub-System
<i>See also:</i>	<i>Originator:</i> Technics (R&D)
<i>Verification method:</i> Test	<i>Verification step:</i> System test
<i>Issues/Decisions:</i>	

EL4-IPS-SSS-REQ-04008-FUR Database Systems for IPS INFRA and SERVICE building block	
IPS shall provide database systems for applications within IPS INFRA and SERVICE building block.	
<i>Comments:</i> One DB system is provided by the INFRA cluster (refer to chapter 6.2.2.11 of SSDD IPS [28]). Two DB systems are provided by the SERVICE cluster (refer to chapter 6.3.2.2 of SSDD IPS [28]).	
<i>Legacy ID:</i> IPS:SSS:REQ:4008	<i>Priority:</i> Mandatory
<i>Status:</i> Approved	<i>Origin:</i> EL4-FL-FEA-0015
<i>Target Release:</i> IPSS OBB 1.0.0 BFZ-3.0	<i>Scope:</i> Sub-System
<i>See also:</i>	<i>Originator:</i> Technics (R&D)
<i>Verification method:</i> Test	<i>Verification step:</i> Integration test
<i>Issues/Decisions:</i>	

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EL4-IPS-SSS-REQ-04009- High Available Database Systems for INFRA and SERVICE building block DPR			
The availability of the database systems within the IPS INFRA and SERVICE building blocks shall be 99.9999%.			
Comments:			
Legacy ID:	IPS:SSS:REQ:4009	Priority:	Mandatory
Status:	Approved	Origin:	EL4-FL-FEA-0015 EL4-DIL-SSS-REQ-00307
Target Release:	IPSS OBB 1.0.0 BFZ-3.0	Scope:	Sub-System
See also:		Originator:	Technics (R&D)
Verification method:	Test	Verification step:	RAM Analysis Endurance test
Issues/Decisions:			

EL4-IPS-SSS-REQ-00041- Feature flags for project specific IPS functionality MCR			
[IPS] shall support using feature flags for configuration, installation and upgrades to allow customization of deployments for various projects using the same [IPS] Target Release.			
Comments: Implement & document feature flags for functionality which might be used only in some projects			
Legacy ID:		Priority:	Mandatory
Status:	Approved	Origin:	EL4-FL-FEA-0015
Target Release:	IPSS OBB 1.0.0	Scope:	Sub-System
See also:		Originator:	Technics (R&D)
Verification method:	Test	Verification step:	Integration test
Issues/Decisions:			

EL4-IPS-SSS-REQ-00042- Dashboards and templates for operations MDR			
[IPS] shall provide pre-configured dashboards and templates for monitoring and alerting to support operational activities.			
Comments:			
Legacy ID:		Priority:	Mandatory
Status:	Approved	Origin:	EL4-FL-FEA-0015
Target Release:	IPSS OBB 1.0.0	Scope:	Sub-System
See also:		Originator:	Technics (R&D)
Verification method:	Test	Verification step:	Integration test
Issues/Decisions:			

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EL4-IPS-SSS-REQ-00045- DPR Minimum Useful Life - Continuous Utilisation	
[IPS] shall guarantee the continuous utilization of the system within the minimum useful life according to [EL4-DIL-SSS-REQ-00300].	
<i>Comments:</i>	
<i>Legacy ID:</i>	<i>Priority:</i> Mandatory
<i>Status:</i> Approved	<i>Origin:</i> EL4-DIL-SSS-REQ-00300
<i>Target Release:</i> IPSS OBB 1.0.0	<i>Scope:</i> Sub-System
<i>See also:</i>	<i>Originator:</i> Technics (R&D)
<i>Verification method:</i> Analysis	<i>Verification step:</i> Analysis
<i>Issues/Decisions:</i>	

EL4-IPS-SSS-REQ-00046- FUR Automatic Self Test - Repeatability	
[IPS] shall be able to execute repeatable self-tests.	
<i>Comments:</i>	
<i>Legacy ID:</i>	<i>Priority:</i> Mandatory
<i>Status:</i> Approved	<i>Origin:</i> EL4-DIL-SSS-REQ-00743 EL4-DIL-SSS-REQ-00744
<i>Target Release:</i> IPSS OBB 1.0.0	<i>Scope:</i> Sub-System
<i>See also:</i>	<i>Originator:</i> Technics (R&D)
<i>Verification method:</i> Test	<i>Verification step:</i> System test
<i>Issues/Decisions:</i>	

EL4-IPS-SSS-REQ-00050- MDR [IPS] - No preventive Maintenance	
[IPS] shall be free of preventive maintenance activities. If this is not technologically feasible for components or if the performance or service life is increased as a result, a maximum of a half-year period maintenance interval is granted.	
<i>Comments:</i>	
<i>Legacy ID:</i>	<i>Priority:</i> Mandatory
<i>Status:</i> Approved	<i>Origin:</i> EL4-DIL-SSS-REQ-00360
<i>Target Release:</i> IPSS OBB 1.0.0	<i>Scope:</i> Sub-System
<i>See also:</i>	<i>Originator:</i> Technics (R&D)
<i>Verification method:</i> Test	<i>Verification step:</i> System test
<i>Issues/Decisions:</i>	

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EL4-IPS-SSS-REQ-00030-MIR		Pre-defined administrative workflows	
[IPS] shall provide pre-defined administrative workflows for rollout and operations.			
<i>Comments:</i> For more details refer to SSS IPS Workflows document [10].			
<i>Legacy ID:</i> <i>Status:</i> Approved		<i>Priority:</i> Mandatory <i>Origin:</i> EL4-DIL-SSS-REQ-00331 EL4-DIL-SSS-REQ-01287 EL4-DIL-SSS-REQ-01290 EL4-DIL-SSS-REQ-01288 EL4-DIL-SSS-REQ-01289 EL4-DIL-SSS-REQ-01302 EL4-DIL-SSS-REQ-01291 EL4-DIL-SSS-REQ-01296 EL4-DIL-SSS-REQ-01295 EL4-DIL-SSS-REQ-01292 EL4-DIL-SSS-REQ-01294 EL4-DIL-SSS-REQ-01298 EL4-DIL-SSS-REQ-01303 EL4-DIL-SSS-REQ-01299 EL4-DIL-SSS-REQ-01300	
<i>Target Release:</i> IPSS OBB 1.0.0 <i>See also:</i> <i>Verification method:</i> Test		<i>Scope:</i> Sub-System <i>Originator:</i> Technics (R&D) <i>Verification step:</i> Integration test	
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EL4-IPS-SSS-REQ-00020-ECR Separation of application and hardware management workflows	
[IPS] shall support independent management workflows of hardware and applications.	
<i>Comments:</i> For more details refer to SSS IPS Workflows document [10].	
<i>Legacy ID:</i> <i>Status:</i> Approved	<i>Priority:</i> Mandatory <i>Origin:</i> EL4-DIL-SSS-REQ-00331 EL4-DIL-SSS-REQ-00568 EL4-DIL-SSS-REQ-01287 EL4-DIL-SSS-REQ-01290 EL4-DIL-SSS-REQ-01288 EL4-DIL-SSS-REQ-01289 EL4-DIL-SSS-REQ-01302 EL4-DIL-SSS-REQ-01291 EL4-DIL-SSS-REQ-01296 EL4-DIL-SSS-REQ-01295 EL4-DIL-SSS-REQ-01292 EL4-DIL-SSS-REQ-01294 EL4-DIL-SSS-REQ-01298 EL4-DIL-SSS-REQ-01303 EL4-DIL-SSS-REQ-01299 EL4-DIL-SSS-REQ-01300
<i>Target Release:</i> IPSS OBB 1.0.0 <i>See also:</i> <i>Verification method:</i> Test	<i>Scope:</i> Sub-System <i>Originator:</i> Technics (R&D) <i>Verification step:</i> System test
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EL4-IPS-SSS-REQ-00035-ECR		Datacenter independent application and hardware management workflows	
[IPS] shall support management workflows of hardware and applications that are independent for each datacenter.			
Comments: For more details refer to SSS IPS Workflows document [10].			
Legacy ID:		Priority:	Mandatory
Status:	Approved	Origin:	EL4-DIL-SSS-REQ-00331 EL4-DIL-SSS-REQ-00568 EL4-DIL-SSS-REQ-01287 EL4-DIL-SSS-REQ-01290 EL4-DIL-SSS-REQ-01288 EL4-DIL-SSS-REQ-01289 EL4-DIL-SSS-REQ-01302 EL4-DIL-SSS-REQ-01291 EL4-DIL-SSS-REQ-01296 EL4-DIL-SSS-REQ-01295 EL4-DIL-SSS-REQ-01292 EL4-DIL-SSS-REQ-01294 EL4-DIL-SSS-REQ-01298 EL4-DIL-SSS-REQ-01303 EL4-DIL-SSS-REQ-01299 EL4-DIL-SSS-REQ-01300
Target Release:	IPSS OBB 2.0.0	Scope:	Sub-System
See also:		Originator:	Technics (R&D)
Verification method:	Test	Verification step:	System test
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EL4-IPS-SSS-REQ-00008-MDR Automated maintenance workflows for geo-redundant applications	
[IPS] shall provide automated maintenance workflows for geo-redundant applications.	
Comments: For more details refer to SSS IPS Workflows document [10].	
Legacy ID: Status: Approved	Priority: Mandatory Origin: EL4-DIL-SSS-REQ-00331 EL4-DIL-SSS-REQ-01287 EL4-DIL-SSS-REQ-01290 EL4-DIL-SSS-REQ-01288 EL4-DIL-SSS-REQ-01289 EL4-DIL-SSS-REQ-01302 EL4-DIL-SSS-REQ-01291 EL4-DIL-SSS-REQ-01296 EL4-DIL-SSS-REQ-01295 EL4-DIL-SSS-REQ-01292 EL4-DIL-SSS-REQ-01294 EL4-DIL-SSS-REQ-01298 EL4-DIL-SSS-REQ-01303 EL4-DIL-SSS-REQ-01299 EL4-DIL-SSS-REQ-01300
Target Release: IPSS OBB 2.0.0	Scope: Sub-System
See also:	Originator: Technics (R&D)
Verification method: Test	Verification step: Integration test
Issues/Decisions:	

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EL4-IPS-SSS-REQ-00009-MDR		Automated maintenance workflows for geo-redundant IPS-related services	
[IPS] shall provide automated maintenance workflows for geo-redundant IPS-related services.			
Comments: For more details refer to SSS IPS Workflows document [10].			
Legacy ID:		Priority:	Mandatory
Status:	Approved	Origin:	EL4-DIL-SSS-REQ-00331 EL4-DIL-SSS-REQ-01287 EL4-DIL-SSS-REQ-01290 EL4-DIL-SSS-REQ-01288 EL4-DIL-SSS-REQ-01289 EL4-DIL-SSS-REQ-01302 EL4-DIL-SSS-REQ-01291 EL4-DIL-SSS-REQ-01296 EL4-DIL-SSS-REQ-01295 EL4-DIL-SSS-REQ-01292 EL4-DIL-SSS-REQ-01294 EL4-DIL-SSS-REQ-01298 EL4-DIL-SSS-REQ-01303 EL4-DIL-SSS-REQ-01299 EL4-DIL-SSS-REQ-01300
Target Release:	IPSS OBB 2.0.0	Scope:	Sub-System
See also:		Originator:	Technics (R&D)
Verification method:	Test	Verification step:	Integration test
Issues/Decisions:			

3.1.2 SPARE NODE MANAGEMENT

EL4-IPS-SSS-REQ-00054-FUR		[IPS] Spare node availability	
[IPS] shall ensure that one spare compute node exist in each Availability Zone for each Host Aggregate.			
Comments:			
Legacy ID:		Priority:	Mandatory
Status:	In negotiation	Origin:	EL4:IDJ-RA:ID:0028 EL4-DIL-SSS-REQ-00357
Target Release:	IPSS OBB 1.0.0	Scope:	Sub-System
See also:		Originator:	Technics (R&D)
Verification method:	Test	Verification step:	System test
Issues/Decisions:	EL4:IDJ-RA:ID:0028 Virtualization Decision: There shall be one compute node spare in each availability zone for each Host Aggregate (but not be part of it). In case of a compute/hypervisor node failure, the spare associated with that aggregate shall become part of it, and the failed resources re-deployed on the spare.		

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EL4-IPS-SSS-REQ-00055- [IPS] Spare node isolation FUR			
The spare compute node shall not be a member of the Host Aggregate during normal operation.			
<i>Comments:</i>			
<i>Legacy ID:</i>		<i>Priority:</i>	Mandatory
<i>Status:</i>	In negotiation	<i>Origin:</i>	EL4:IDJ-RA:ID:0028 EL4-DIL-SSS-REQ-00357
<i>Target Release:</i>	IPSS OBB 1.0.0	<i>Scope:</i>	Sub-System
<i>See also:</i>		<i>Originator:</i>	Technics (R&D)
<i>Verification method:</i>	Test	<i>Verification step:</i>	System test
<i>Issues/Decisions:</i>	EL4:IDJ-RA:ID:0028 Virtualization Decision: There shall be one compute node spare in each availability zone for each Host Aggregate (but not be part of it). In case of a compute/hypervisor node failure, the spare associated with that aggregate shall become part of it, and the failed resources re-deployed on the spare.		

EL4-IPS-SSS-REQ-00056- [IPS] Spare node activation FUR			
In case of compute / hypervisor node failure [IPS] shall add the corresponding spare compute node to the affected Host Aggregate			
<i>Comments:</i>			
<i>Legacy ID:</i>		<i>Priority:</i>	Mandatory
<i>Status:</i>	In negotiation	<i>Origin:</i>	EL4:IDJ-RA:ID:0028 EL4-DIL-SSS-REQ-00357
<i>Target Release:</i>	IPSS OBB 1.0.0	<i>Scope:</i>	Sub-System
<i>See also:</i>		<i>Originator:</i>	Technics (R&D)
<i>Verification method:</i>	Test	<i>Verification step:</i>	System test
<i>Issues/Decisions:</i>	EL4:IDJ-RA:ID:0028 Virtualization Decision: There shall be one compute node spare in each availability zone for each Host Aggregate (but not be part of it). In case of a compute/hypervisor node failure, the spare associated with that aggregate shall become part of it, and the failed resources re-deployed on the spare.		

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EL4-IPS-SSS-REQ-00057- [IPS] Spare reprovisioning FUR			
[IPS] shall re-deploy the workloads from the failed node in the spare compute node.			
<i>Comments:</i>			
<i>Legacy ID:</i>		<i>Priority:</i>	Mandatory
<i>Status:</i>	In negotiation	<i>Origin:</i>	EL4:IDJ-RA:ID:0028 EL4-DIL-SSS-REQ-00357
<i>Target Release:</i>	IPSS OBB 1.0.0	<i>Scope:</i>	Sub-System
<i>See also:</i>		<i>Originator:</i>	Technics (R&D)
<i>Verification method:</i>	Test	<i>Verification step:</i>	System test
<i>Issues/Decisions:</i>	EL4:IDJ-RA:ID:0028 Virtualization Decision: There shall be one compute node spare in each availability zone for each Host Aggregate (but not be part of it). In case of a compute/hypervisor node failure, the spare associated with that aggregate shall become part of it, and the failed resources re-deployed on the spare.		

3.1.3 OEVE/OENORM AND EULYNX

EL4-IPS-SSS-REQ-00051- Fulfillment of Standard [OEVE/OENORM] FUR			
The requirements according to [OEVE/OENORM EN 50126], [OEVE/OENORM EN 50716], [OEVE/OENORM EN 50129] shall be fulfilled.			
<i>Comments:</i>			
<i>Legacy ID:</i>		<i>Priority:</i>	Mandatory
<i>Status:</i>	Approved	<i>Origin:</i>	EL4-DIL-SSS-REQ-00303 EL4-DIL-SSS-REQ-00304 EL4-DIL-SSS-REQ-00305
<i>Target Release:</i>	IPSS OBB 1.0.0	<i>Scope:</i>	Sub-System
<i>See also:</i>		<i>Originator:</i>	Technics (R&D)
<i>Verification method:</i>	Review	<i>Verification step:</i>	Review
<i>Issues/Decisions:</i>			

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EL4-IPS-SSS-REQ-00052-FUR		Fulfillment of EULYNX Standard Eu.Doc.16	
The requirements according to [EULYNX Specification] - (Eu.Doc.16) shall be fulfilled.			
Comments:			
Legacy ID:		Priority:	Mandatory
Status:	Approved	Origin:	EL4-DIL-SSS-REQ-00306
Target Release:	IPSS OBB 1.0.0	Scope:	Sub-System
See also:		Originator:	Technics (R&D)
Verification method:	Review	Verification step:	Review
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3.1.4 MONITORING SYSTEM [FRACAS]

EL4-IPS-SSS-REQ-00053- [FRACAS] - Data provision FUR	
[IPS] shall provide data inputs required by [FRACAS].	
<i>Comments:</i> [IPS] provides data to [FRACAS] based in accordance with the applicable IPS requirements: •[EL4-IPS-SSS-REQ-01005] - Diagnostic Network Data •[EL4-IPS-SSS-REQ-04002] - Monitoring Stack for OpenStack •[EL4-IPS-SSS-REQ-04003] - Log Collection Stack for OpenStack •[EL4-IPS-SSS-REQ-05201] - Monitoring Stack for INFRA Cluster •[EL4-IPS-SSS-REQ-05202] - Log Collection Stack for INFRA Cluster •[EL4-IPS-SSS-REQ-06303] - Monitoring Stack for SERVICE Cluster •[EL4-IPS-SSS-REQ-06304] - Log Collection Stack for SERVICE Cluster •[EL4-IPS-SSS-REQ-07500] - Functional Data Retrieval of Application and System Log Data for IPS INFRA, SERVICE, NETWORK MONITORING and BASE building block •[EL4-IPS-SSS-REQ-08005] - Power Supply Interface: Monitoring of USVs	
<i>Legacy ID:</i> <i>Status:</i> Approved <i>Target Release:</i> IPSS OBB 1.0.0 <i>See also:</i> <i>Verification method:</i> Test	<i>Priority:</i> Mandatory <i>Origin:</i> EL4-DIL-SSS-REQ-00672 EL4-DIL-SSS-REQ-00673 EL4-DIL-SSS-REQ-00674 EL4-DIL-SSS-REQ-00675 EL4-DIL-SSS-REQ-00676 EL4-DIL-SSS-REQ-00678 EL4-DIL-SSS-REQ-00679 EL4-DIL-SSS-REQ-00680 EL4-DIL-SSS-REQ-00681 <i>Scope:</i> Sub-System <i>Originator:</i> Technics (R&D) <i>Verification step:</i> System test
<i>Issues/Decisions:</i>	

3.2 IPS BASE

IPS BASE provides the base layer for networking, computing and storage.

3.2.1 CORE NETWORK

Highly available network within one location Datacenter.

Refer for more details to SSDD Core Network Design Description [24].

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3.2.1.1 NETWORK DEVICES

Requirements regarding local network devices that shall be connected to the Core Network. This includes all cabinets and additional devices like power supply supervision.

EL4-IPS-SSS-REQ-01001- Server Cabinets PER			
The Core Network shall support connectivity to a minimum of three server cabinets			
<i>Comments:</i> Refer for more details to SSDD Generic Server and Network Cabinet [12] and SSDD Core Network Design Description [24]. Scalability function shall be considered according to [EL4-IPS-SSS-REQ-04031] - [EL4-IPS-SSS-REQ-04033]			
<i>Legacy ID:</i>		<i>Priority:</i>	Mandatory
<i>Status:</i>	Approved	<i>Origin:</i>	EL4-FL-FEA-0015
<i>Target Release:</i>	IPSS OBB 1.0.0	<i>Scope:</i>	Sub-System
<i>See also:</i>	EL4-IPS-SSS-REQ-04031 EL4-IPS-SSS-REQ-04032 EL4-IPS-SSS-REQ-04033	<i>Originator:</i>	Technics (R&D)
<i>Verification method:</i>	Test	<i>Verification step:</i>	Review
<i>Issues/Decisions:</i>			

EL4-IPS-SSS-REQ-01005- Diagnostic Network Data MDR			
The following diagnostic network data shall be provided by using the NETWORK MONITORING tooling: •Jitter •Response Time •Packet Loss •Connectivity •Round Trip Time			
<i>Comments:</i> The NSP tool provides diagnostic data by creation of reports. For more information refer to Chapter 6.5 of SSDD IPS [28].			
<i>Legacy ID:</i>	IPS:SSS:REQ:1005	<i>Priority:</i>	Mandatory
<i>Status:</i>	Approved	<i>Origin:</i>	EL4-FL-FEA-0015 EL4-DIL-SSS-REQ-00330 EL4-DIL-SSS-REQ-00329
<i>Target Release:</i>	IPSS OBB 1.0.0 BFZ-3.0	<i>Scope:</i>	Sub-System
<i>See also:</i>		<i>Originator:</i>	Technics (R&D)
<i>Verification method:</i>	Test	<i>Verification step:</i>	System test
<i>Issues/Decisions:</i>			

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3.2.2 EXTERNAL NETWORK CONNECTIONS

These networks connect external systems to IPS. There are several different connection types specified as own requirements.

EL4-IPS-SSS-REQ-01102- Power Supply: physical connection DPR			
The Core Network shall be able to provide two-fold redundant connections to the Power Supply Monitor device from Gustav Klein.			
<i>Comments:</i>			
<i>Legacy ID:</i>	IPS:SSS:REQ:1102	<i>Priority:</i>	Mandatory
<i>Status:</i>	Approved	<i>Origin:</i>	EL4-FL-FEA-0015 EL4-DIL-SSS-REQ-00326 EL4-DIL-SSS-REQ-00328
<i>Target Release:</i>	IPSS OBB 1.0.0 BFZ-3.0	<i>Scope:</i>	Sub-System
<i>See also:</i>		<i>Originator:</i>	Technics (R&D)
<i>Verification method:</i>	Test	<i>Verification step:</i>	System test
<i>Issues/Decisions:</i>			

EL4-IPS-SSS-REQ-01103- OEGB Network DPR			
The Core Network shall be able to provide two-fold redundant connections to the OEGB network.			
<i>Comments:</i> OEGB network: SiDiNET. For more detail refer to SSDD Core Network Design Description [24].			
<i>Legacy ID:</i>	IPS:SSS:REQ:1103	<i>Priority:</i>	Mandatory
<i>Status:</i>	Approved	<i>Origin:</i>	EL4-DIL-SSS-REQ-01142 EL4-DIL-SSS-REQ-01137 EL4-DIL-SSS-REQ-00326 EL4-DIL-SSS-REQ-00328
<i>Target Release:</i>	IPSS OBB 1.0.0 BFZ-3.0	<i>Scope:</i>	Sub-System
<i>See also:</i>		<i>Originator:</i>	Technics (R&D)
<i>Verification method:</i>	Test	<i>Verification step:</i>	System test
<i>Issues/Decisions:</i>			

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EL4-IPS-SSS-REQ-01104- RaSTA Connections FUR			
The Core Network shall support redundant RaSTA connections.			
Comments:			
Legacy ID:		Priority:	Mandatory
Status:	Approved	Origin:	EL4-DIL-SSS-REQ-00697 EL4-DIL-SSS-REQ-00464 EL4-DIL-SSS-REQ-00465 EL4-DIL-SSS-REQ-00466 EL4-DIL-SSS-REQ-00326 EL4-DIL-SSS-REQ-00328 EL4-DIL-SSS-REQ-01473 EL4-DIL-SSS-REQ-00732
Target Release:	IPSS OBB 1.0.0	Scope:	Sub-System
See also:		Originator:	Technics (R&D)
Verification method:	Test	Verification step:	System test
Issues/Decisions:			

EL4-IPS-SSS-REQ-01105- IP address for the [ÖBB] [NTP] time server CSR			
IP address shall be configurable via IPS without the need for a new baseline release.			
Comments:			
Legacy ID:		Priority:	Mandatory
Status:	Approved	Origin:	EL4-DIL-SSS-REQ-00557
Target Release:	IPSS OBB 1.0.0	Scope:	Sub-System
See also:		Originator:	Technics (R&D)
Verification method:	Test	Verification step:	System test
Issues/Decisions:			

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EL4-IPS-SSS-REQ-01106- NTP traffic markers FUR	
[IPS] shall ensure that NTP traffic to the [ÖBB] time server is marked with the appropriate traffic marker for the interface specified by EULYNX or EU-Rail SP documents.	
<i>Comments:</i>	
<i>Legacy ID:</i>	<i>Priority:</i> Mandatory
<i>Status:</i> Approved	<i>Origin:</i> EL4-DIL-SSS-REQ-00547
<i>Target Release:</i> IPSS OBB 1.0.0	<i>Scope:</i> Sub-System
<i>See also:</i> EL4-IPS-SSS-REQ-90026 EL4-IPS-SSS-REQ-90025	<i>Originator:</i> Technics (R&D)
<i>Verification method:</i> Test	<i>Verification step:</i> System test
<i>Issues/Decisions:</i>	

3.2.3 INTERCONNECTION NETWORK

This chapter covers all Interconnection LAN related requirements.

EL4-IPS-SSS-REQ-01201- Connection to customer Interconnection LAN FUR	
The Core Network shall be connected to the customers Interconnection LAN via two independent lines with a bandwidth of 50Gbit/sec per line.	
<i>Comments:</i>	
<i>Legacy ID:</i>	<i>Priority:</i> Mandatory
<i>Status:</i> Approved	<i>Origin:</i> EL4-DIL-SSS-REQ-00551 EL4-DIL-SSS-REQ-00326 EL4-DIL-SSS-REQ-00328
<i>Target Release:</i> IPSS OBB 1.0.0	<i>Scope:</i> Sub-System
<i>See also:</i>	<i>Originator:</i> Technics (R&D)
<i>Verification method:</i> Test	<i>Verification step:</i> System test
<i>Issues/Decisions:</i>	

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EL4-IPS-SSS-REQ-01202-CSR [DC] locations network condition: delay	
In addition to REQ [EL4-IPS-SSS-REQ-01201] the following conditions for the network between the [DC] locations shall be fulfilled. Maximum latency between sender and recipient: 15 ms	
<i>Comments:</i>	
<i>Legacy ID:</i>	<i>Priority:</i> Mandatory
<i>Status:</i> Approved	<i>Origin:</i> EL4-DIL-SSS-REQ-00551
<i>Target Release:</i> IPSS OBB 1.0.0	<i>Scope:</i> Sub-System
<i>See also:</i>	<i>Originator:</i> Technics (R&D)
<i>Verification method:</i> Test	<i>Verification step:</i> System test
<i>Issues/Decisions:</i>	

EL4-IPS-SSS-REQ-01203-CSR [DC] locations network condition: Permissible jitter between redundant lines	
In addition to REQ [EL4-IPS-SSS-REQ-01201] the following conditions for the network between the [DC] locations shall be fulfilled. Permissible jitter between redundant lines: 4 ms	
<i>Comments:</i>	
<i>Legacy ID:</i>	<i>Priority:</i> Mandatory
<i>Status:</i> Approved	<i>Origin:</i> EL4-DIL-SSS-REQ-00551
<i>Target Release:</i> IPSS OBB 1.0.0	<i>Scope:</i> Sub-System
<i>See also:</i>	<i>Originator:</i> Technics (R&D)
<i>Verification method:</i> Test	<i>Verification step:</i> System test
<i>Issues/Decisions:</i>	

EL4-IPS-SSS-REQ-01204-CSR [DC] locations network condition: delay: Permissible jitter between sender and receiver	
In addition to REQ [EL4-IPS-SSS-REQ-01201] the following conditions for the network between the [DC] locations shall be fulfilled. Permissible jitter between sender and receiver: 4.5 ms (30%)	
<i>Comments:</i>	
<i>Legacy ID:</i>	<i>Priority:</i> Mandatory
<i>Status:</i> Approved	<i>Origin:</i> EL4-DIL-SSS-REQ-00551
<i>Target Release:</i> IPSS OBB 1.0.0	<i>Scope:</i> Sub-System
<i>See also:</i>	<i>Originator:</i> Technics (R&D)
<i>Verification method:</i> Test	<i>Verification step:</i> System test
<i>Issues/Decisions:</i>	

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EL4-IPS-SSS-REQ-01205-CSR [DC] locations network condition: delay: Permissible bit error rate (BER)	
In addition to REQ [EL4-IPS-SSS-REQ-01201] the following conditions for the network between the [DC] locations shall be fulfilled. Permissible bit error rate per redundant network: BER = 1E-13.	
<i>Comments:</i>	
<i>Legacy ID:</i>	<i>Priority:</i> Mandatory
<i>Status:</i> Approved	<i>Origin:</i> EL4-DIL-SSS-REQ-00551
<i>Target Release:</i> IPSS OBB 1.0.0	<i>Scope:</i> Sub-System
<i>See also:</i>	<i>Originator:</i> Technics (R&D)
<i>Verification method:</i> Test	<i>Verification step:</i> System test
<i>Issues/Decisions:</i>	

3.2.4 DEPENDABILITY REQUIREMENTS - NETWORK

Dependability requirements covering various reliability, availability, maintainability requirements for the networking.

EL4-IPS-SSS-REQ-01301-DPR Availability of Core Network	
The Core Network shall provide 99.9999% availability.	
<i>Comments:</i> For the configuration and architecture refer to SSDD Core Network Design Description [24] and the network design as specified by the supplier according to DDD Low Level Design Network document [08].	
<i>Legacy ID:</i> IPS:SSS:REQ:1301	<i>Priority:</i> Mandatory
<i>Status:</i> Approved	<i>Origin:</i> EL4-FL-FEA-0015
<i>Target Release:</i> IPSS OBB 1.0.0	<i>Scope:</i> Sub-System
<i>See also:</i>	<i>Originator:</i> Technics (R&D)
<i>Verification method:</i> Review	<i>Verification step:</i> RAM Analysis Endurance test
<i>Issues/Decisions:</i>	

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EL4-IPS-SSS-REQ-01302-FUR		Separation of maintenance and operational (non-safety / safety) network	
Maintenance and operational (non-safety / safety) network shall be separated.			
<i>Comments:</i> Current maintenance networks are (list can change in the future): "mgmt-vpls" "lnz-maas-ha" "sbg-maas-ha" "wls-maas-ha" For all currently configured maintenance networks and networks for non-safety and safety related operations refer to SSDD Core Network Design Description [24].			
<i>Legacy ID:</i>	IPS:SSS:REQ:1302	<i>Priority:</i>	Mandatory
<i>Status:</i>	Approved	<i>Origin:</i>	EL4-FL-FEA-0015
<i>Target Release:</i>	IPSS OBB 1.0.0 BFZ-3.0	<i>Scope:</i>	Sub-System
<i>See also:</i>		<i>Originator:</i>	Technics (R&D)
<i>Verification method:</i>	Review	<i>Verification step:</i>	Review
<i>Issues/Decisions:</i>			

EL4-IPS-SSS-REQ-01304-DPR		Server Cabinets: two-fold redundant network connection	
The network connection to each server cabinet shall be two-fold redundant.			
Comments: For more information refer to SSDD Core Network Design Description [24].			
Legacy ID:	IPS:SSS:REQ:1304	Priority:	Mandatory
Status:	Approved	Origin:	EL4-DIL-SSS-REQ-01135 EL4-DIL-SSS-REQ-00326 EL4-DIL-SSS-REQ-00328 EL4-DIL-SSS-REQ-01473
Target Release:	IPSS OBB 1.0.0 BFZ-3.0	Scope:	Sub-System
See also:		Originator:	Technics (R&D)
Verification method:	Test	Verification step:	Review
Issues/Decisions:			

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EL4-IPS-SSS-REQ-01306- Two-fold redundant network connection to RaSTA networks DPR			
The network connection to each RaSTA network shall be two-fold redundant.			
<i>Comments:</i> For more details related RaSTA interfaces refer to document [07] and to Chapter 4.1.2 of SSDD Core Network Design Description [24].			
<i>Legacy ID:</i>	IPS:SSS:REQ:1306	<i>Priority:</i>	Mandatory
<i>Status:</i>	Approved	<i>Origin:</i>	EL4-DIL-SSS-REQ-00989 EL4-DIL-SSS-REQ-00326 EL4-DIL-SSS-REQ-00328 EL4-DIL-SSS-REQ-01473
<i>Target Release:</i>	IPSS OBB 1.0.0 BFZ-3.0	<i>Scope:</i>	Sub-System
<i>See also:</i>		<i>Originator:</i>	Technics (R&D)
<i>Verification method:</i>	Test	<i>Verification step:</i>	Review
<i>Issues/Decisions:</i>			

EL4-IPS-SSS-REQ-01307- Redundancy in network topology DPR			
The connection or interfaces of network components shall be designed according to the Network Design Description [24] to mitigate a single point of failure.			
<i>Comments:</i> For more details refer to SSDD OMC 3.x Core Network Design Description [24].			
<i>Legacy ID:</i>	IPS:SSS:REQ:1307	<i>Priority:</i>	Mandatory
<i>Status:</i>	Approved	<i>Origin:</i>	EL4-DIL-SSS-REQ-00326 EL4-DIL-SSS-REQ-00328 EL4-DIL-SSS-REQ-01473
<i>Target Release:</i>	IPSS OBB 1.0.0 BFZ-3.0	<i>Scope:</i>	Sub-System
<i>See also:</i>		<i>Originator:</i>	Technics (R&D)
<i>Verification method:</i>	Test	<i>Verification step:</i>	Review
<i>Issues/Decisions:</i>			

3.2.5 STORAGE SYSTEM

IPS uses an open-source software-defined storage platform for high-redundant local data storage.

3.2.5.1 DEPENDABILITY REQUIREMENTS - STORAGE SYSTEM

Dependability requirements covering various reliability, availability, maintainability requirements for the storage system.

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EL4-IPS-SSS-REQ-02001- Storage System Availability DPR			
The volumes provided by the storage system shall provide 99.9999% availability.			
<i>Comments:</i>			
<i>Legacy ID:</i>	IPS:SSS:REQ:2001	<i>Priority:</i>	Mandatory
<i>Status:</i>	Approved	<i>Origin:</i>	EL4-FL-FEA-0015 EL4-DIL-SSS-REQ-00406
<i>Target Release:</i>	IPSS OBB 1.0.0 BFZ-3.0	<i>Scope:</i>	Sub-System
<i>See also:</i>		<i>Originator:</i>	Technics (R&D)
<i>Verification method:</i>	Review	<i>Verification step:</i>	RAM Analysis Endurance test
<i>Issues/Decisions:</i>			

EL4-IPS-SSS-REQ-02002- Operating System RAID on Storage Servers DPR			
The operating system on all storage servers shall use at least 2 storage devices in a RAID1 array.			
<i>Comments:</i> The concrete type of file system is ext4. Refer to more information to Installation Guide IPS [32].			
<i>Legacy ID:</i>	IPS:SSS:REQ:2002	<i>Priority:</i>	Mandatory
<i>Status:</i>	Approved	<i>Origin:</i>	EL4-FL-FEA-0015
<i>Target Release:</i>	IPSS OBB 1.0.0 BFZ-3.0	<i>Scope:</i>	Sub-System
<i>See also:</i>		<i>Originator:</i>	Technics (R&D)
<i>Verification method:</i>	Test	<i>Verification step:</i>	Review
<i>Issues/Decisions:</i>			

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EL4-IPS-SSS-REQ-02003- HW-Disk redundancy for storage servers DPR			
The storage server shall provide additional storage devices for the provision of the common storage space for the applications.			
<i>Comments:</i> Regarding the number of hard disks and performance refer to requirement GENCAB:HW-SSS:REQ:0006 of Generic Server and Network Cabinet SSS [06]. The concrete type of file system is ext4.			
<i>Legacy ID:</i>	IPS:SSS:REQ:2003	<i>Priority:</i>	Mandatory
<i>Status:</i>	Approved	<i>Origin:</i>	EL4-FL-FEA-0015
<i>Target Release:</i>	IPSS OBB 1.0.0 BFZ-3.0	<i>Scope:</i>	Sub-System
<i>See also:</i>		<i>Originator:</i>	Technics (R&D)
<i>Verification method:</i>	Test	<i>Verification step:</i>	Review
<i>Issues/Decisions:</i>			

EL4-IPS-SSS-REQ-02004- Block storage for stateful applications FUR			
The storage system shall provide a block storage for stateful applications.			
<i>Comments:</i> To provide persistent storage to stateful application containers, Kubernetes can use services from the underlying IaaS layer. This means, Kubernetes will request a dedicated block storage device from the IaaS layer to be attached to the worker node the application container was scheduled to and take care of mounting the requested volume into the container's filesystem namespace.			
<i>Legacy ID:</i>	IPS:SSS:REQ:2004	<i>Priority:</i>	Mandatory
<i>Status:</i>	Approved	<i>Origin:</i>	EL4-FL-FEA-0015
<i>Target Release:</i>	IPSS OBB 1.0.0 BFZ-3.0	<i>Scope:</i>	Sub-System
<i>See also:</i>		<i>Originator:</i>	Technics (R&D)
<i>Verification method:</i>	Test	<i>Verification step:</i>	Part System Test
<i>Issues/Decisions:</i>			

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EL4-IPS-SSS-REQ-02005- Object storage for stateful applications FUR			
The storage system shall provide an object storage for stateful applications.			
<i>Comments:</i> Object storage, also known as object-based storage, is a computer data storage architecture designed to handle large amounts of unstructured data. Unlike other architectures, it designates data as distinct units, bundled with metadata and a unique identifier that can be used to locate and access each data unit.			
<i>Legacy ID:</i>	IPS:SSS:REQ:2005	<i>Priority:</i>	Mandatory
<i>Status:</i>	Approved	<i>Origin:</i>	EL4-FL-FEA-0015 EL4-DIL-SSS-REQ-01182
<i>Target Release:</i>	IPSS OBB 1.0.0 BFZ-3.0	<i>Scope:</i>	Sub-System
<i>See also:</i>		<i>Originator:</i>	Technics (R&D)
<i>Verification method:</i>	Test	<i>Verification step:</i>	Part System Test
<i>Issues/Decisions:</i>			

3.2.6 COMPUTING

IPS uses 33 servers per [Datacenter] acting as compute nodes (11 in each server cabinet defined in the Generic Server and Network Cabinet SSS [06]) for running business applications on it to operate, administrate and monitor instances and the location itself.

3.2.6.1 DEPENDABILITY REQUIREMENTS - COMPUTING

Dependability requirements covering various reliability, availability, maintainability requirements for computing.

EL4-IPS-SSS-REQ-03101- Operating System RAID on Compute Server DPR			
The operating system on all compute servers shall use at least 2 storage devices in a RAID1 array.			
<i>Comments:</i> The concrete type of file system is ext4. Refer to more information to Installation Guide IPS [32].			
<i>Legacy ID:</i>	IPS:SSS:REQ:3101	<i>Priority:</i>	Mandatory
<i>Status:</i>	Approved	<i>Origin:</i>	EL4-FL-FEA-0015
<i>Target Release:</i>	IPSS OBB 1.0.0 BFZ-3.0	<i>Scope:</i>	Sub-System
<i>See also:</i>		<i>Originator:</i>	Technics (R&D)
<i>Verification method:</i>	Test	<i>Verification step:</i>	Review
<i>Issues/Decisions:</i>			

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3.2.7 IPS BASE PROVIDER SOFTWARE

For each location one OpenStack instance is created with the following requirements. This instance includes OpenStack projects for INFRA Kubernetes Cluster.

3.2.7.1 GENERIC IPS BASE FUNCTIONS

This chapter gathers all global functions shared by all necessary clusters.

EL4-IPS-SSS-REQ-04001- NTP Service FUR			
Each NTP service on the OpenStack control plane shall able to connect to NTP servers provided by the customer and shall sync. their time with each other. All compute and storage servers shall sync. their time with the NTP services on the OpenStack control plane.			
<i>Comments:</i> The control plane node of OpenStack will provide an internal NTP-Service and is connected to NTP-servers provided by the customer. Refer for more details to Chapter 6.2.8 of SSDD IPS [28].			
<i>Legacy ID:</i>	IPS:SSS:REQ:4001	<i>Priority:</i>	Mandatory
<i>Status:</i>	Approved	<i>Origin:</i>	EL4-DIL-SSS-REQ-00560
<i>Target Release:</i>	IPSS OBB 1.0.0 BFZ-3.0	<i>Scope:</i>	Sub-System
<i>See also:</i>	EL4-IPS-SSS-REQ-90025 EL4-IPS-SSS-REQ-90026	<i>Originator:</i>	Technics (R&D)
<i>Verification method:</i>	Test	<i>Verification step:</i>	Review System test
<i>Issues/Decisions:</i>			

EL4-IPS-SSS-REQ-04002- Monitoring Stack for OpenStack MDR			
The OpenStack instance shall provide a monitoring stack to monitor hardware on the location.			
<i>Comments:</i> The monitoring stack shall be deployed on each control plane VM (3 control plane VMs per location).			
<i>Legacy ID:</i>	IPS:SSS:REQ:4002	<i>Priority:</i>	Mandatory
<i>Status:</i>	Approved	<i>Origin:</i>	EL4-FL-FEA-0015
<i>Target Release:</i>	IPSS OBB 1.0.0 BFZ-3.0	<i>Scope:</i>	Sub-System
<i>See also:</i>		<i>Originator:</i>	Technics (R&D)
<i>Verification method:</i>	Test	<i>Verification step:</i>	System test
<i>Issues/Decisions:</i>			

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EL4-IPS-SSS-REQ-04003- Log Collection Stack for OpenStack MDR			
The OpenStack instance shall provide a log collection stack to collect system log data on OpenStack level, accessible only to authorized persons with read-only permissions.			
<i>Comments:</i> The log collection stack shall be deployed on each control plane VM (3 control plane VMs per location) and provides means to analyze system log data files.			
<i>Legacy ID:</i>	IPS:SSS:REQ:4003	<i>Priority:</i>	Mandatory
<i>Status:</i>	Approved	<i>Origin:</i>	EL4-FL-FEA-0015
<i>Target Release:</i>	IPSS OBB 1.0.0 BFZ-3.0	<i>Scope:</i>	Sub-System
<i>See also:</i>		<i>Originator:</i>	Technics (R&D)
<i>Verification method:</i>	Test	<i>Verification step:</i>	System test
<i>Issues/Decisions:</i>			

EL4-IPS-SSS-REQ-04015- Storage and computing services DPR			
[IPS] shall provide high available storage and computing services spanning 3 availability zones			
<i>Comments:</i>			
<i>Legacy ID:</i>		<i>Priority:</i>	Mandatory
<i>Status:</i>	Approved	<i>Origin:</i>	EL4:IDJ-RA:ID:0005 EL4-DIL-SSS-REQ-01483 EL4-DIL-SSS-REQ-01484
<i>Target Release:</i>	IPSS OBB 1.0.0	<i>Scope:</i>	Sub-System
<i>See also:</i>		<i>Originator:</i>	Technics (R&D)
<i>Verification method:</i>	Test	<i>Verification step:</i>	Integration test
<i>Issues/Decisions:</i>	EL4:IDJ-RA:ID:0005 Consistent data within and across [Datacenter]s Decision: IPS needs to guarantee that data is consistent across all Availability Zones and datacenters.		

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EL4-IPS-SSS-REQ-04017- PER Separation of resources	
[IPS] shall provide physical separation of storage, networking and computing HW resources.	
<i>Comments:</i>	
<i>Legacy ID:</i> <i>Status:</i> Approved <i>Target Release:</i> IPSS OBB 1.0.0 <i>See also:</i> <i>Verification method:</i> Review	<i>Priority:</i> Mandatory <i>Origin:</i> EL4-FL-FEA-0015 EL4-DIL-SSS-REQ-00217 EL4-DIL-SSS-REQ-01483 EL4-DIL-SSS-REQ-01484 <i>Scope:</i> Sub-System <i>Originator:</i> Technics (R&D) <i>Verification step:</i> Review
<i>Issues/Decisions:</i>	

EL4-IPS-SSS-REQ-04018- PER Support of redundant resources	
[IPS] shall support redundant power-supply (PDU), storage, networking and computing HW resources.	
<i>Comments:</i>	
<i>Legacy ID:</i> <i>Status:</i> Approved <i>Target Release:</i> IPSS OBB 1.0.0 <i>See also:</i> <i>Verification method:</i> Review	<i>Priority:</i> Mandatory <i>Origin:</i> EL4-DIL-SSS-REQ-00324 EL4-DIL-SSS-REQ-00325 <i>Scope:</i> Sub-System <i>Originator:</i> Technics (R&D) <i>Verification step:</i> Review
<i>Issues/Decisions:</i>	

EL4-IPS-SSS-REQ-04019- OAR Multitenancy for VM workloads	
[IPS] shall provide multitenancy of VM workloads.	
<i>Comments:</i>	
<i>Legacy ID:</i> <i>Status:</i> Approved <i>Target Release:</i> IPSS OBB 1.0.0 <i>See also:</i> <i>Verification method:</i> Test	<i>Priority:</i> Mandatory <i>Origin:</i> EL4:IDJ-RA:ID:0004 <i>Scope:</i> Sub-System <i>Originator:</i> Technics (R&D) <i>Verification step:</i> Integration test
<i>Issues/Decisions:</i> EL4:IDJ-RA:ID:0004 Workload separation based on OpenStack Decision: Usage of OpenStack concepts in IPS for managing VM and workload separation.	

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EL4-IPS-SSS-REQ-04020- OAR Separation of concerns for VM workloads	
[IPS] shall provide workload isolation for VM workloads.	
<i>Comments:</i>	
<i>Legacy ID:</i>	<i>Priority:</i> Mandatory
<i>Status:</i> Approved	<i>Origin:</i> EL4:IDJ-RA:ID:0004 EL4-DIL-SSS-REQ-00217
<i>Target Release:</i> IPSS OBB 1.0.0	<i>Scope:</i> Sub-System
<i>See also:</i>	<i>Originator:</i> Technics (R&D)
<i>Verification method:</i> Test	<i>Verification step:</i> Integration test
<i>Issues/Decisions:</i> EL4:IDJ-RA:ID:0004 Workload separation based on OpenStack Decision: Usage of OpenStack concepts in IPS for managing VM and workload separation.	

EL4-IPS-SSS-REQ-04021- FUR Network technologies	
[IPS] shall provide [MPLS], [RSVP], [VPLS] and [LSP] local network configurations.	
<i>Comments:</i>	
<i>Legacy ID:</i>	<i>Priority:</i> Mandatory
<i>Status:</i> Approved	<i>Origin:</i> EL4-DIL-SSS-REQ-00546 EL4-DIL-SSS-REQ-00548 EL4-DIL-SSS-REQ-00545
<i>Target Release:</i> IPSS OBB 1.0.0	<i>Scope:</i> Sub-System
<i>See also:</i> [14] Chapter 1.3.4	<i>Originator:</i> Technics (R&D)
<i>Verification method:</i> Test	<i>Verification step:</i> System test
<i>Issues/Decisions:</i>	

EL4-IPS-SSS-REQ-04022- FUR Different IP segments	
Physically separated networks and their [PoS] signaling interfaces shall use different IP segments supported by the network technologies specified in the [EL4-IPS-SSS-REQ-04021].	
<i>Comments:</i>	
<i>Legacy ID:</i>	<i>Priority:</i> Mandatory
<i>Status:</i> Approved	<i>Origin:</i> EL4-DIL-SSS-REQ-00546
<i>Target Release:</i> IPSS OBB 1.0.0	<i>Scope:</i> Sub-System
<i>See also:</i> [14] Chapter 1.3.4	<i>Originator:</i> Technics (R&D)
<i>Verification method:</i> Test	<i>Verification step:</i> System test
<i>Issues/Decisions:</i>	

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EL4-IPS-SSS-REQ-04023- Physical network configuration FUR	
[IPS] shall ensure consistent, up-to-date configuration of network hardware.	
<i>Comments:</i>	
<i>Legacy ID:</i>	<i>Priority:</i> Mandatory
<i>Status:</i> Approved	<i>Origin:</i> EL4-DIL-SSS-REQ-00546
<i>Target Release:</i> IPSS OBB 1.0.0	<i>Scope:</i> Sub-System
<i>See also:</i> [14] Chapter 1.3.4	<i>Originator:</i> Technics (R&D)
<i>Verification method:</i> Test	<i>Verification step:</i> System test
<i>Issues/Decisions:</i>	

EL4-IPS-SSS-REQ-04024- Configuration capabilities for VLANs FUR	
[IPS] shall support VLANs on the operating system and network devices.	
<i>Comments:</i>	
<i>Legacy ID:</i>	<i>Priority:</i> Mandatory
<i>Status:</i> Approved	<i>Origin:</i>
<i>Target Release:</i> IPSS OBB 1.0.0	<i>Scope:</i> Sub-System
<i>See also:</i> [14] Chapter 1.3.4 EL4-IPS-SSS-REQ-08306	<i>Originator:</i> Technics (R&D)
<i>Verification method:</i> Test	<i>Verification step:</i> System test
<i>Issues/Decisions:</i>	

EL4-IPS-SSS-REQ-04025- Support of local Availability Zone concept DPR	
[IPS] shall support local Availability Zone concept for VMs.	
<i>Comments:</i>	
<i>Legacy ID:</i>	<i>Priority:</i> Mandatory
<i>Status:</i> Approved	<i>Origin:</i> EL4-DIL-SSS-REQ-00450 EL4-DIL-SSS-REQ-00451
<i>Target Release:</i> IPSS OBB 1.0.0	<i>Scope:</i> Sub-System
<i>See also:</i>	<i>Originator:</i> Technics (R&D)
<i>Verification method:</i> Test	<i>Verification step:</i> System test
<i>Issues/Decisions:</i>	

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EL4-IPS-SSS-REQ-04026- Support of Geo-Redundancy concept DPR	
[IPS] shall support Geo-Redundancy (multiple regions) concept for VMs.	
<i>Comments:</i>	
<i>Legacy ID:</i>	<i>Priority:</i> Mandatory
<i>Status:</i> Approved	<i>Origin:</i> EL4-DIL-SSS-REQ-00458
<i>Target Release:</i> IPSS OBB 2.0.0	<i>Scope:</i> Sub-System
<i>See also:</i>	<i>Originator:</i> Technics (R&D)
<i>Verification method:</i> Test	<i>Verification step:</i> System test
<i>Issues/Decisions:</i>	

EL4-IPS-SSS-REQ-04027- IPv4 for network layer OSI Layer FUR	
[IPS] shall ensure that VMs and network interfaces provisioned in the network layer OSI Layer 3 are configured to use IPv4.	
<i>Comments:</i>	
<i>Legacy ID:</i>	<i>Priority:</i> Mandatory
<i>Status:</i> Approved	<i>Origin:</i> EL4-DIL-SSS-REQ-00549
<i>Target Release:</i> IPSS OBB 1.0.0	<i>Scope:</i> Sub-System
<i>See also:</i>	<i>Originator:</i> Technics (R&D)
<i>Verification method:</i> Test	<i>Verification step:</i> System test
<i>Issues/Decisions:</i>	

EL4-IPS-SSS-REQ-04028- Ability to manage linux VMs FUR	
[IPS] shall provide the ability to manage Linux VMs.	
<i>Comments:</i> Ability to manage Linux VMs includes capabilities to create, configure, start, stop, manage the lifecycle of virtual Machines including monitoring their statuses and adjusting resources (CPU, memory, storage) if required.	
<i>Legacy ID:</i>	<i>Priority:</i> Mandatory
<i>Status:</i> Approved	<i>Origin:</i> EL4-FL-FEA-0015
<i>Target Release:</i> IPSS OBB 1.0.0	<i>Scope:</i> Sub-System
<i>See also:</i>	<i>Originator:</i> Technics (R&D)
<i>Verification method:</i> Test	<i>Verification step:</i> Integration test
<i>Issues/Decisions:</i>	

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EL4-IPS-SSS-REQ-04029-Ability to manage VM images			
FUR			
[IPS] shall provide the ability to manage VM images.			
Comments: Ability to manage VM images for VMs includes capabilities to create, update and delete images ensuring consistent and repeatable VMs deployment.			
Legacy ID:		Priority:	Mandatory
Status:	Approved	Origin:	EL4-DIL-SSS-REQ-00573 EL4-DIL-SSS-REQ-00818
Target Release:	IPSS OBB 1.0.0	Scope:	Sub-System
See also:		Originator:	Technics (R&D)
Verification method:	Test	Verification step:	Integration test
Issues/Decisions:			

EL4-IPS-SSS-REQ-04030-DPR				Automatic restart of the VMs			
[IPS] shall support automatic restart of the VMs on the server after the server restart.							
Comments:							
Legacy ID:				Priority: Mandatory			
Status: Approved				Origin: EL4-DIL-SSS-REQ-00655			
Target Release: IPSS OBB 1.0.0				Scope: Sub-System			
See also:				Originator: Technics (R&D)			
Verification method: Test				Verification step: System test			
Issues/Decisions:							

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EL4-IPS-SSS-REQ-04031- Additional Computing HW Resources DPR	
[IPS] shall support scale-in / out (increase or decrease) additional computing resources per Availability zone.	
Comments:	
Legacy ID: Status: Approved Target Release: IPSS OBB 1.0.0 See also: Verification method: Test	Priority: Mandatory Origin: EL4:IDJ-RA:ID:0006 EL4-DIL-SSS-REQ-01483 EL4-DIL-SSS-REQ-01484 Scope: Sub-System Originator: Technics (R&D) Verification step: Integration test Performance test
Issues/Decisions: EL4:IDJ-RA:ID:0006 Horizontal scaling of IPS and hardware cabinets across availability zones (AZ) Decision: IPS shall support increased resource demands (storage and/or compute) and scale horizontally per Availability zone.	

EL4-IPS-SSS-REQ-04032- Additional Network HW Resources DPR	
[IPS] shall support scale-in / out (increase or decrease) additional network resources per Availability zone.	
Comments:	
Legacy ID: Status: Approved Target Release: IPSS OBB 1.0.0 See also: Verification method: Test	Priority: Mandatory Origin: EL4:IDJ-RA:ID:0006 Scope: Sub-System Originator: Technics (R&D) Verification step: Integration test Performance test
Issues/Decisions: EL4:IDJ-RA:ID:0006 Horizontal scaling of IPS and hardware cabinets across availability zones (AZ) Decision: IPS shall support increased resource demands (storage and/or compute) and scale horizontally per Availability zone.	

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EL4-IPS-SSS-REQ-04033- DPR Additional Storage HW Resources	
[IPS] shall support scale-in / out (increase or decrease) additional storage resources per Availability zone.	
<i>Comments:</i>	
<i>Legacy ID:</i> <i>Status:</i> Approved <i>Target Release:</i> IPSS OBB 1.0.0 <i>See also:</i> <i>Verification method:</i> Test	<i>Priority:</i> Mandatory <i>Origin:</i> EL4:IDJ-RA:ID:0006 <i>Scope:</i> Sub-System <i>Originator:</i> Technics (R&D) <i>Verification step:</i> Integration test Performance test
<i>Issues/Decisions:</i> EL4:IDJ-RA:ID:0006 Horizontal scaling of IPS and hardware cabinets across availability zones (AZ) Decision: IPS shall support increased resource demands (storage and/or compute) and scale horizontally per Availability zone.	

EL4-IPS-SSS-REQ-04034- FUR IP segments for physical separation	
[IPS] shall assign distinct non-overlapping IP segments to physically separate networks and their PoS signalling interfaces, ensuring strict isolation	
<i>Comments:</i>	
<i>Legacy ID:</i> <i>Status:</i> In negotiation <i>Target Release:</i> IPSS OBB 1.0.0 <i>See also:</i> <i>Verification method:</i> Test	<i>Priority:</i> Mandatory <i>Origin:</i> EL4-DIL-SSS-REQ-00546 <i>Scope:</i> Sub-System <i>Originator:</i> Technics (R&D) <i>Verification step:</i> System test
<i>Issues/Decisions:</i>	

EL4-IPS-SSS-REQ-04035- FUR Provision of PHC	
IPS shall support virtual PHC for NMR VMs	
<i>Comments:</i>	
<i>Legacy ID:</i> <i>Status:</i> Approved <i>Target Release:</i> IPSS OBB 1.0.0 <i>See also:</i> [43]PLF-2.x:SSRS:NMR:0122 <i>Verification method:</i> Test	<i>Priority:</i> Mandatory <i>Origin:</i> EL4:IDJ-RA:ID:0010 <i>Scope:</i> Sub-System <i>Originator:</i> Technics (R&D) <i>Verification step:</i> Integration test
<i>Issues/Decisions:</i> EL4:IDJ-RA:ID:0010 IPS-VIL time synchronization Decision: IPS provides the system time to VIL instances via an emulated PHC clock.	

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EL4-IPS-SSS-REQ-04036- Provision of Operating Systems FUR	
[IPS] Base shall provide Operating Systems for the Servers.	
<i>Comments:</i>	
<i>Legacy ID:</i>	<i>Priority:</i> Mandatory
<i>Status:</i> Approved	<i>Origin:</i> EL4-DIL-SSS-REQ-00567
<i>Target Release:</i> IPSS OBB 1.0.0	<i>Scope:</i> Sub-System
<i>See also:</i>	<i>Originator:</i> Technics (R&D)
<i>Verification method:</i> Test	<i>Verification step:</i> Integration test
<i>Issues/Decisions:</i>	

EL4-IPS-SSS-REQ-04037- Provision of CMOS data storage FUR	
IPS shall support consistent Stateful Application Data Storage for Elektra Core	
<i>Comments:</i>	
<i>Legacy ID:</i>	<i>Priority:</i> Mandatory
<i>Status:</i> Approved	<i>Origin:</i> EL4:IDJ-RA:ID:0010 EL4:IDJ-RA:ID:0011 EL4:IDJ-RA:ID:0012
<i>Target Release:</i> IPSS OBB 1.0.0	<i>Scope:</i> Sub-System
<i>See also:</i>	<i>Originator:</i> Technics (R&D)
<i>Verification method:</i> Test	<i>Verification step:</i> Integration test
<i>Issues/Decisions:</i> EL4:IDJ-RA:ID:0010 IPS-VIL time synchronization Decision: IPS provides the system time to VIL instances via an emulated PHC clock. EL4:IDJ-RA:ID:0011 VIL-IPS CMOS data storage Decision: For the duration of a complete VIL station shutdown (e.g. planned maintenance, software upgrades, etc.), CMOS data shall be persisted by IPS. EL4:IDJ-RA:ID:0012 Stateful Application Data Storage for Elektra Core Decision: IPS shall provide the interface IF-IPS-01 (Stateful Application Data Storage) to VIL.	

3.3 IPS INFRA

The INFRA cluster contains containerized applications to administrate and manage SERVICE clusters and provides infrastructure services related to the location.

3.3.1 OPENSTACK PROJECT FOR INFRA CLUSTER

This chapter covers functional requirements for OpenStack project related to the INFRA clusters.

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EL4-IPS-SSS-REQ-04101- Virtual Machines for INFRA Cluster FUR			
The INFRA OpenStack project shall manage up to 10 VMs.			
<i>Comments:</i> Total amount of nodes: 3 control plane VMs + 6 worker VMs + 1 Jumpbox VM = 10 VMs			
<i>Legacy ID:</i>	IPS:SSS:REQ:4101	<i>Priority:</i>	Mandatory
<i>Status:</i>	Approved	<i>Origin:</i>	EL4-FL-FEA-0015
<i>Target Release:</i>	IPSS OBB 1.0.0 BFZ-3.0	<i>Scope:</i>	Sub-System
<i>See also:</i>		<i>Originator:</i>	Technics (R&D)
<i>Verification method:</i>	Test	<i>Verification step:</i>	Part System Test
<i>Issues/Decisions:</i>			

EL4-IPS-SSS-REQ-04102- Virtual Machines for containerized non-safety-related applications on INFRA Cluster FUR			
The INFRA OpenStack project shall manage VMs to run containerized non-safety-related applications.			
<i>Comments:</i>			
<i>Legacy ID:</i>	IPS:SSS:REQ:4102	<i>Priority:</i>	Mandatory
<i>Status:</i>	Approved	<i>Origin:</i>	EL4-FL-FEA-0015
<i>Target Release:</i>	IPSS OBB 1.0.0 BFZ-3.0	<i>Scope:</i>	Sub-System
<i>See also:</i>		<i>Originator:</i>	Technics (R&D)
<i>Verification method:</i>	Test	<i>Verification step:</i>	Part System Test
<i>Issues/Decisions:</i>			

EL4-IPS-SSS-REQ-04103- Jumpbox Virtual Machine for INFRA and SERVICE cluster rollout/maintenance FUR			
The INFRA OpenStack project shall manage a VM used to rollout or maintain the INFRA cluster and both SERVICE clusters.			
<i>Comments:</i> For more details refer to chapter 6.2.3 of SSDD IPS document [28].			
<i>Legacy ID:</i>	IPS:SSS:REQ:4103	<i>Priority:</i>	Mandatory
<i>Status:</i>	Approved	<i>Origin:</i>	EL4-FL-FEA-0015 EL4-DIL-SSS-REQ-00568
<i>Target Release:</i>	IPSS OBB 1.0.0 BFZ-3.0	<i>Scope:</i>	Sub-System
<i>See also:</i>		<i>Originator:</i>	Technics (R&D)
<i>Verification method:</i>	Test	<i>Verification step:</i>	Integration test
<i>Issues/Decisions:</i>			

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3.3.2 NON-SAFETY-RELATED CONTAINER REQUIREMENTS FOR INFRA CLUSTER

This chapter covers all non-safety-related application container requirements for the INFRA cluster.

EL4-IPS-SSS-REQ-05001- Containerized non-safety-related Applications on INFRA Cluster FUR			
The INFRA cluster shall be able to run containerized non-safety-related applications running under dedicated system accounts.			
<i>Comments:</i>			
<i>Legacy ID:</i>	IPS:SSS:REQ:5001	<i>Priority:</i>	Mandatory
<i>Status:</i>	Approved	<i>Origin:</i>	EL4-FL-FEA-0015
<i>Target Release:</i>	IPSS OBB 1.0.0 BFZ-3.0	<i>Scope:</i>	Sub-System
<i>See also:</i>		<i>Originator:</i>	Technics (R&D)
<i>Verification method:</i>	Test	<i>Verification step:</i>	Integration test
<i>Issues/Decisions:</i>			

3.3.3 SHARED SERVICES (INFRA) FOR INFRA CLUSTER

This chapter covers all requirements of using shared services which are necessary to operate management and monitoring functions.

EL4-IPS-SSS-REQ-05104- High Available Storage for Application Files DPR			
The INFRA cluster shall provide a storage for application files with an availability of 99.9999%.			
<i>Comments:</i>			
<i>Legacy ID:</i>	IPS:SSS:REQ:5104	<i>Priority:</i>	Mandatory
<i>Status:</i>	Approved	<i>Origin:</i>	EL4-FL-FEA-0015
<i>Target Release:</i>	IPSS OBB 1.0.0 BFZ-3.0	<i>Scope:</i>	Sub-System
<i>See also:</i>		<i>Originator:</i>	Technics (R&D)
<i>Verification method:</i>	Test	<i>Verification step:</i>	RAM Analysis Endurance test
<i>Issues/Decisions:</i>			

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EL4-IPS-SSS-REQ-05105- High Available Storage for Data Preparation Files DPR			
The INFRA cluster shall provide a storage for data preparation files with an availability of 99.9999%.			
<i>Comments:</i>			
<i>Legacy ID:</i>	IPS:SSS:REQ:5105	<i>Priority:</i>	Mandatory
<i>Status:</i>	Approved	<i>Origin:</i>	EL4-FL-FEA-0015
<i>Target Release:</i>	IPSS OBB 1.0.0 BFZ-3.0	<i>Scope:</i>	Sub-System
<i>See also:</i>		<i>Originator:</i>	Technics (R&D)
<i>Verification method:</i>	Test	<i>Verification step:</i>	RAM Analysis Endurance test
<i>Issues/Decisions:</i>			

EL4-IPS-SSS-REQ-05106- High Available Storage for Configuration Files DPR			
The INFRA cluster shall provide a storage for configuration files with an availability of 99.9999%.			
<i>Comments:</i>			
<i>Legacy ID:</i>	IPS:SSS:REQ:5106	<i>Priority:</i>	Mandatory
<i>Status:</i>	Approved	<i>Origin:</i>	EL4-FL-FEA-0015
<i>Target Release:</i>	IPSS OBB 1.0.0 BFZ-3.0	<i>Scope:</i>	Sub-System
<i>See also:</i>		<i>Originator:</i>	Technics (R&D)
<i>Verification method:</i>	Test	<i>Verification step:</i>	RAM Analysis Endurance test
<i>Issues/Decisions:</i>			

EL4-IPS-SSS-REQ-05107- Global Central Repository for Storing Artefacts FUR			
The INFRA cluster shall provide a global central repository for storing artefacts.			
<i>Comments:</i> This artefact storage shall be used for storing artefacts like security patches or OS images, etc. and doesn't to need to be high available, because storage is used only during retrofit scenario procedures (not during normal operation).			
<i>Legacy ID:</i>	IPS:SSS:REQ:5107	<i>Priority:</i>	Mandatory
<i>Status:</i>	Approved	<i>Origin:</i>	EL4-FL-FEA-0015
<i>Target Release:</i>	IPSS OBB 1.0.0 BFZ-3.0	<i>Scope:</i>	Sub-System
<i>See also:</i>		<i>Originator:</i>	Technics (R&D)
<i>Verification method:</i>	Test	<i>Verification step:</i>	Integration test
<i>Issues/Decisions:</i>			

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3.3.4 SHARED SERVICES (GDP) FOR INFRA CLUSTER

This chapter covers all requirements of using shared GDP services which are necessary to operate management and monitoring functions.

EL4-IPS-SSS-REQ-05201- Monitoring Stack for INFRA Cluster MDR			
The INFRA cluster shall provide a monitoring stack to monitor applications running on the INFRA cluster.			
<i>Comments:</i> For the list of monitored components refer to SDD Monitoring Concept [25] and chapter 6.3.2.8 of SSDD IPS [28].			
<i>Legacy ID:</i>	IPS:SSS:REQ:5201	<i>Priority:</i>	Mandatory
<i>Status:</i>	Approved	<i>Origin:</i>	EL4-FL-FEA-0015
<i>Target Release:</i>	IPSS OBB 1.0.0 BFZ-3.0	<i>Scope:</i>	Sub-System
<i>See also:</i>		<i>Originator:</i>	Technics (R&D)
<i>Verification method:</i>	Test	<i>Verification step:</i>	Integration test
<i>Issues/Decisions:</i>			

EL4-IPS-SSS-REQ-05202- Log Collection Stack for INFRA Cluster MDR			
The INFRA cluster shall provide a log collection stack to collect application and system log data within the INFRA cluster, accessible only to authorized persons with read-only permissions.			
<i>Comments:</i> This log collection stack provides means to analyze application and system log data files. The stack provides the following features to secure and protect the retrieved log data: Role-Based Access Control (RBAC) feature: ensures users and services to grant only the minimum level of access required to perform their specific tasks within the stack. Refer to SDD Users and Roles Management [29] and chapter 6.1.1.4 of SSDD IPS [28] document Unchangeable of log entries: omit possibility of changing log data content			
For more details about logging refer to SDD Logging Concept [27] and chapter 6.3.2.9 of SSDD IPS [28] document.			
<i>Legacy ID:</i>	IPS:SSS:REQ:5202	<i>Priority:</i>	Mandatory
<i>Status:</i>	Approved	<i>Origin:</i>	EL4-FL-FEA-0015
<i>Target Release:</i>	IPSS OBB 1.0.0 BFZ-3.0	<i>Scope:</i>	Sub-System
<i>See also:</i>		<i>Originator:</i>	Technics (R&D)
<i>Verification method:</i>	Test	<i>Verification step:</i>	Security Approval Integration test
<i>Issues/Decisions:</i>			

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EL4-IPS-SSS-REQ-05203- Provision of Backup facility for INFRA Cluster			
FUR			
A backup service shall be provided for log collection stack (including stack configuration and log data) and database system for the INFRA cluster.			
<i>Comments:</i> For more details regarding backup refer to chapter 6.3.2.10 of SSDD IPS [28] document.			
<i>Legacy ID:</i>	IPS:SSS:REQ:5203	<i>Priority:</i>	Mandatory
<i>Status:</i>	Approved	<i>Origin:</i>	EL4-FL-FEA-0015
<i>Target Release:</i>	IPSS OBB 1.0.0 BFZ-3.0	<i>Scope:</i>	Sub-System
<i>See also:</i>		<i>Originator:</i>	Technics (R&D)
<i>Verification method:</i>	Test	<i>Verification step:</i>	Integration test
<i>Issues/Decisions:</i>			

EL4-IPS-SSS-REQ-05302- PKI Service for Management of Certificates			
SER			
The INFRA cluster shall provide public key infrastructure (PKI) for management of certificates.			
<i>Comments:</i> All communication between Kubernetes nodes, web-interfaces, etc. are encrypted. Therefore, certificates are needed. The PKI service manages (generates and rotates) certificates on demand. This allows applications to acquire certificates without going through the usual manual process of generating a private key and Certificate Signing Request (CSR), submitting to a Certificate Authority (CA), and then waiting for the verification and signing process to complete.			
<i>Legacy ID:</i>	IPS:SSS:REQ:5302	<i>Priority:</i>	Mandatory
<i>Status:</i>	Approved	<i>Origin:</i>	EL4-FL-FEA-0015
<i>Target Release:</i>	IPSS OBB 1.0.0 BFZ-3.0	<i>Scope:</i>	Sub-System
<i>See also:</i>		<i>Originator:</i>	Technics (R&D)
<i>Verification method:</i>	Test	<i>Verification step:</i>	Integration test System test
<i>Issues/Decisions:</i>			

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EL4-IPS-SSS-REQ-05306- DPR Availability of the PKI Service for Management of Certificates			
The availability of the PKI service for management of certificates shall be 99.9999%.			
Comments:			
Legacy ID:	IPS:SSS:REQ:5306	Priority:	Mandatory
Status:	Approved	Origin:	EL4-FL-FEA-0015
Target Release:	IPSS OBB 1.0.0 BFZ-3.0	Scope:	Sub-System
See also:		Originator:	Technics (R&D)
Verification method:	Test	Verification step:	RAM Analysis Endurance test
Issues/Decisions:			

3.4 IPS SERVICE

3.4.1 NON-SAFETY-RELATED CONTAINER REQUIREMENTS FOR SERVICE CLUSTER

EL4-IPS-SSS-REQ-06101- FUR Containerized Non-safety-related Applications on SERVICE Cluster			
The SERVICE cluster shall able to run containerized non-safety-related applications running under dedicated system accounts.			
Comments:			
Legacy ID:	IPS:SSS:REQ:6101	Priority:	Mandatory
Status:	Approved	Origin:	EL4-FL-FEA-0015
Target Release:	IPSS OBB 1.0.0 BFZ-3.0	Scope:	Sub-System
See also:		Originator:	Technics (R&D)
Verification method:	Test	Verification step:	Integration test
Issues/Decisions:			

3.4.2 SHARED SERVICES (GDP) FOR SERVICE CLUSTER

This chapter covers all requirements of using shared GDP services which are necessary to operate functions.

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EL4-IPS-SSS-REQ-06302- Message Broker for SERVICE Cluster FUR			
The SERVICE cluster shall provide a message broker for communication between business components.			
Comments:			
Legacy ID:	IPS:SSS:REQ:6302	Priority:	Mandatory
Status:	Approved	Origin:	EL4-FL-FEA-0015
Target Release:	IPSS OBB 1.0.0 BFZ-3.0	Scope:	Sub-System
See also:		Originator:	Technics (R&D)
Verification method:	Test	Verification step:	Integration test
Issues/Decisions:			

EL4-IPS-SSS-REQ-06303- Monitoring Stack for SERVICE Cluster MDR			
The SERVICE cluster shall provide a monitoring stack to monitor applications running on the SERVICE cluster.			
Comments: For the list of monitored components refer to SDD Monitoring Concept [25] and chapter 6.4.2.3 of SSDD IPS [24].			
Legacy ID:	IPS:SSS:REQ:6303	Priority:	Mandatory
Status:	Approved	Origin:	EL4-FL-FEA-0015
Target Release:	IPSS OBB 1.0.0 BFZ-3.0	Scope:	Sub-System
See also:		Originator:	Technics (R&D)
Verification method:	Test	Verification step:	Integration test
Issues/Decisions:			

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EL4-IPS-SSS-REQ-06304- MDR Log Collection Stack for SERVICE Cluster			
The SERVICE cluster shall provide a log collection stack to collect application and system log data, accessible only to authorized persons with read-only permissions.			
<i>Comments:</i> This log collection stack provides means to analyze application and system log data files. The stack provides the features to secure and protect the retrieved log data. The following ones are just some of them: Role-Based Access Control (RBAC) feature: ensures users and services to grant only the minimum level of access required to perform their specific tasks within the stack. Refer to SDD Users and Roles Management [29] and chapter 6.1.1.4 of SSDD IPS [28] document Unchangeable of log entries: omit possibility of changing log data content For more details about logging refer to SDD Logging Concept [27] and chapter 6.4.2.4 of SSDD IPS [28] document.			
<i>Legacy ID:</i>	IPS:SSS:REQ:6304	<i>Priority:</i>	Mandatory
<i>Status:</i>	Approved	<i>Origin:</i>	EL4-FL-FEA-0015
<i>Target Release:</i>	IPSS OBB 1.0.0 BFZ-3.0	<i>Scope:</i>	Sub-System
<i>See also:</i>		<i>Originator:</i>	Technics (R&D)
<i>Verification method:</i>	Test	<i>Verification step:</i>	Integration test
<i>Issues/Decisions:</i>			

EL4-IPS-SSS-REQ-06305- FUR Provision of Backup facility for SERVICE Cluster			
A backup service shall be provided for log collection stack (including stack configuration and log data) and the two database systems of the SERVICE cluster.			
<i>Comments:</i> For more details regarding backup refer to chapter 6.4.2.7 of SSDD IPS [28] document.			
<i>Legacy ID:</i>	IPS:SSS:REQ:6305	<i>Priority:</i>	Mandatory
<i>Status:</i>	Approved	<i>Origin:</i>	EL4-FL-FEA-0015
<i>Target Release:</i>	IPSS OBB 1.0.0 BFZ-3.0	<i>Scope:</i>	Sub-System
<i>See also:</i>		<i>Originator:</i>	Technics (R&D)
<i>Verification method:</i>	Test	<i>Verification step:</i>	System test
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EL4-IPS-SSS-REQ-06306- DPR Availability of the Message Broker			
The availability of the message broker shall be 99.9999%.			
<i>Comments:</i>			
<i>Legacy ID:</i>	IPS:SSS:REQ:6306	<i>Priority:</i>	Mandatory
<i>Status:</i>	Approved	<i>Origin:</i>	EL4-FL-FEA-0015
<i>Target Release:</i>	IPSS OBB 1.0.0 BFZ-3.0	<i>Scope:</i>	Sub-System
<i>See also:</i>		<i>Originator:</i>	Technics (R&D)
<i>Verification method:</i>	Test	<i>Verification step:</i>	RAM Analysis Endurance test
<i>Issues/Decisions:</i>			

3.5 IPS NETWORK MONITORING

The NSP tool (software provided by NOKIA) is an administration and monitoring tool for NOKIA networks. The NSP is a distributed system that provides comprehensive infrastructure, service management, and resource control within and among the whole Interconnection Network.

3.5.1 OPENSTACK PROJECT FOR NETWORK MONITORING

This chapter covers functional requirements for OpenStack project related to the NETWORK MONITORING instance of each location.

EL4-IPS-SSS-REQ-07001- FUR Virtual Machines for NETWORK MONITORING tool (service)			
The NETWORK MONITORING OpenStack project shall manage up to 3 VMs.			
<i>Comments:</i> The following 3 VMs have to be provided by IPS to run the NETWORK MONITORING tool (instance) from NOKIA: •Deployer-Node-VM •NFM-P-collocated-VM •NSP-Cluster-VM			
<i>Legacy ID:</i>	IPS:SSS:REQ:7001	<i>Priority:</i>	Mandatory
<i>Status:</i>	Approved	<i>Origin:</i>	EL4-FL-FEA-0015
<i>Target Release:</i>	IPSS OBB 1.0.0 BFZ-3.0	<i>Scope:</i>	Sub-System
<i>See also:</i>		<i>Originator:</i>	Technics (R&D)
<i>Verification method:</i>	Test	<i>Verification step:</i>	System test
<i>Issues/Decisions:</i>			

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EL4-IPS-SSS-REQ-07002-FUR		Connection to NETWORK MONITORING instance running on the other location	
IPS shall provide the connection to the NETWORK MONITORING instance running on the other location using the interconnection LAN.			
<i>Comments:</i> Based on the geo-redundant implementation across locations within the NETWORK MONITORING tool (one instance is in active and the other one is in standby mode) both NETWORK MONITORING instances shall be connected together. For more information refer to the network design as specified by the supplier according to DDD Low Level Design Network document [08].			
<i>Legacy ID:</i>	IPS:SSS:REQ:7002	<i>Priority:</i>	Mandatory
<i>Status:</i>	Approved	<i>Origin:</i>	EL4-FL-FEA-0015
<i>Target Release:</i>	IPSS OBB 1.0.0 BFZ-3.0	<i>Scope:</i>	Sub-System
<i>See also:</i>		<i>Originator:</i>	Technics (R&D)
<i>Verification method:</i>	Test	<i>Verification step:</i>	System test
<i>Issues/Decisions:</i>			

EL4-IPS-SSS-REQ-07003-FUR		Connection to all network components for the NETWORK MONITORING instance	
IPS shall provide the connection to all network components within the Core Network and interconnection LAN for the NETWORK MONITORING instance to administer and monitor those components.			
<i>Comments:</i> To administer and monitor all network components within the Core Network and Interconnection Network from the NETWORK MONITORING instance the connectivity to all the network components must be established.			
For more information refer to the network design as specified by the supplier according to DDD Low Level Design Network document [08].			
<i>Legacy ID:</i>	IPS:SSS:REQ:7003	<i>Priority:</i>	Mandatory
<i>Status:</i>	Approved	<i>Origin:</i>	EL4-FL-FEA-0015
<i>Target Release:</i>	IPSS OBB 1.0.0 BFZ-3.0	<i>Scope:</i>	Sub-System
<i>See also:</i>		<i>Originator:</i>	Technics (R&D)
<i>Verification method:</i>	Test	<i>Verification step:</i>	System test
<i>Issues/Decisions:</i>			

3.6 IPS ROLLOUT, MAINTENANCE AND EMERGENCY

This chapter covers requirements regarding first IPS rollout, access to IPS maintenance procedures in including data repatriation in German "Datenheimholung" and in case of emergency situations access to the system.

IPS maintenance procedures are provided and implemented as IPS Workflows. They are documented in SSS IPS Workflows [10], SSDD IPS Workflows [33] and SDD IPS Workflows [34]. Here is IPS shared

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service and it provides maintenance tools for all software and hardware components distributed over all IPS building blocks: INFRA, SERVICE, NETWORK MONITORING and BASE.

EL4-IPS-SSS-REQ-07500-MDR		Functional Data Retrieval of Application and System Log Data for IPS INFRA, SERVICE, NETWORK MONITORING and BASE building block	
IPS shall provide an export functionality Data Repatriation of: •collected application and system log data from all log collection stack components •collected NSP network analysis reports User role with read-only privileges will be used during the data repatriation procedure.			
<i>Comments:</i> The data repatriation process starts via a cron job on the “Jumpbox Data Repatriation VM” (refer to EL4-IPS-SSS-REQ-07503) to collect data from the DSTW system. Current procedure foresees only for authorized persons, dedicated technical system account and tools on a read-only basis: Retrieving – depending on the amount of log data – on daily and/or hourly interval snapshot files from the central log storages containing all application and system log data. Retrieving NSP network analysis reports (refer to EL4-IPS-SSS-REQ-01005) from the NSP tool Create an encrypted data repatriation transfer packet containing all files and transfer it to the administration workstation (here ends the responsibility of IPS) Exact definition refer to SDD Logging Concept document [27].			
<i>Legacy ID:</i>	IPS:SSS:REQ:7500	<i>Priority:</i>	Mandatory
<i>Status:</i>	Approved	<i>Origin:</i>	EL4-FL-FEA-0015
<i>Target Release:</i>	IPSS OBB 1.0.0 BFZ-3.0	<i>Scope:</i>	Sub-System
<i>See also:</i>		<i>Originator:</i>	Technics (R&D)
<i>Verification method:</i>	Test	<i>Verification step:</i>	Integration test System test
<i>Issues/Decisions:</i>			

3.6.1 OPENSTACK PROJECT “ADMIN”

This chapter covers functional requirements for IaaS OpenStack project “Admin”.

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EL4-IPS-SSS-REQ-07501- Virtual Machine for IPS Rollout FUR			
The “Admin” OpenStack project shall manage a VM (Kayobe VM) providing IPS rollout procedures.			
<i>Comments:</i> IPS rollout procedures are triggered by: •the Hitachi Rail Service Center team with privileged local accounts from one of the administrative workstations only as defined in Rollout Concept document [40] and SSDD IPS document [28] •a dedicated service account during retrofit scenario execution as defined in SDD IPS Workflows [34]			
<i>Legacy ID:</i>	IPS:SSS:REQ:7501	<i>Priority:</i>	Mandatory
<i>Status:</i>	Approved	<i>Origin:</i>	EL4-FL-FEA-0015
<i>Target Release:</i>	IPSS OBB 1.0.0 BFZ-3.0	<i>Scope:</i>	Sub-System
<i>See also:</i>		<i>Originator:</i>	Technics (R&D)
<i>Verification method:</i>	Test	<i>Verification step:</i>	Integration test
<i>Issues/Decisions:</i>			

EL4-IPS-SSS-REQ-07502- Virtual Machine for Maintenance and Emergency Activities FUR			
The “Admin” OpenStack project shall manage a VM used for maintenance and emergency activities.			
<i>Comments:</i> Maintenance and emergency activities are triggered by: •the Hitachi Rail Service Center team with privileged local accounts from one of the administrative workstations only as defined in SDD Users and Roles Managements and SSDD IPS document [28] •a dedicated service account during retrofit scenario execution as defined in SDD IPS Workflows [34]			
<i>Legacy ID:</i>	IPS:SSS:REQ:7502	<i>Priority:</i>	Mandatory
<i>Status:</i>	Approved	<i>Origin:</i>	EL4-FL-FEA-0015 EL4-DIL-SSS-REQ-00568
<i>Target Release:</i>	IPSS OBB 1.0.0 BFZ-3.0	<i>Scope:</i>	Sub-System
<i>See also:</i>		<i>Originator:</i>	Technics (R&D)
<i>Verification method:</i>	Test	<i>Verification step:</i>	Integration test
<i>Issues/Decisions:</i>			

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EL4-IPS-SSS-REQ-07503- Jumpbox Virtual Machine for Data Repatriation FUR			
The “Admin” OpenStack project shall manage a VM used for the data repatriation process specified in SDD Logging Concept document [28].			
<i>Comments:</i> For more details refer to SDD Logging Concept document [28].			
<i>Legacy ID:</i>	IPS:SSS:REQ:7503	<i>Priority:</i>	Mandatory
<i>Status:</i>	Approved	<i>Origin:</i>	EL4-FL-FEA-0015
<i>Target Release:</i>	IPSS OBB 1.0.0 BFZ-3.0	<i>Scope:</i>	Sub-System
<i>See also:</i>		<i>Originator:</i>	Technics (R&D)
<i>Verification method:</i>	Test	<i>Verification step:</i>	Integration test
<i>Issues/Decisions:</i>			

3.7 EXTERNAL INTERFACES

All external interface requirements are listed here.

EL4-IPS-SSS-REQ-08002- Interface to OEGB Datennetz 2020 FUR			
IPS shall provide the connection to the OEGB network Datennetz 2020 for relevant interfaces (listed within comment).			
<i>Comments:</i> The following interfaces shall be connected to the OEGB network Datennetz 2020: IF-49 ADM User Interface			
<i>Legacy ID:</i>	IPS:SSS:REQ:8002	<i>Priority:</i>	Mandatory
<i>Status:</i>	Approved	<i>Origin:</i>	EL4-DIL-SSS-REQ-01142 EL4-DIL-SSS-REQ-01137
<i>Target Release:</i>	IPSS OBB 1.0.0 BFZ-3.0	<i>Scope:</i>	Sub-System
<i>See also:</i>		<i>Originator:</i>	Technics (R&D)
<i>Verification method:</i>	Test	<i>Verification step:</i>	System test
<i>Issues/Decisions:</i>			

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EL4-IPS-SSS-REQ-08005- Power Supply Interface: Monitoring of USVs FUR			
IPS shall monitor UPSs via the Power Supply Monitor device from Gustav Klein.			
<i>Comments:</i> The interface IF-25 is specified in document [37].			
<i>Legacy ID:</i>	IPS:SSS:REQ:8005	<i>Priority:</i>	Mandatory
<i>Status:</i>	Approved	<i>Origin:</i>	EL4-FL-FEA-0015
<i>Target Release:</i>	IPSS OBB 1.0.0 BFZ-3.0	<i>Scope:</i>	Sub-System
<i>See also:</i>		<i>Originator:</i>	Technics (R&D)
<i>Verification method:</i>	Test	<i>Verification step:</i>	System test
<i>Issues/Decisions:</i>			

EL4-IPS-SSS-REQ-08006- Interface to OMC - X25 over RaSTA FUR			
IPS shall provide connectivity to [OMC] system.			
<i>Comments:</i> The communication protocol is [X25 over RaSTA].			
<i>Legacy ID:</i>		<i>Priority:</i>	Mandatory
<i>Status:</i>	Approved	<i>Origin:</i>	EL4-DIL-SSS-REQ-00464
<i>Target Release:</i>	IPSS OBB 1.0.0	<i>Scope:</i>	Sub-System
<i>See also:</i>	[15] IPS:SSS:REQ:8001	<i>Originator:</i>	Technics (R&D)
<i>Verification method:</i>	Test	<i>Verification step:</i>	System test
<i>Issues/Decisions:</i>			

EL4-IPS-SSS-REQ-08007- Interface to SCWS - X25 over RaSTA FUR			
IPS shall provide connectivity to [SCWS] system.			
<i>Comments:</i> The communication protocol is [X25 over RaSTA].			
<i>Legacy ID:</i>		<i>Priority:</i>	Mandatory
<i>Status:</i>	Approved	<i>Origin:</i>	EL4-DIL-SSS-REQ-00470
<i>Target Release:</i>	IPSS OBB 1.0.0	<i>Scope:</i>	Sub-System
<i>See also:</i>	[15] IPS:SSS:REQ:8001	<i>Originator:</i>	Technics (R&D)
<i>Verification method:</i>	Test	<i>Verification step:</i>	System test
<i>Issues/Decisions:</i>			

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EL4-IPS-SSS-REQ-08008- Interface to RBC - X25 over RaSTA FUR			
IPS shall provide connectivity to [RBC] system.			
<i>Comments:</i> The communication protocol is [X25 over RaSTA].			
<i>Legacy ID:</i>		<i>Priority:</i>	Mandatory
<i>Status:</i>	Approved	<i>Origin:</i>	EL4-DIL-SSS-REQ-00468
<i>Target Release:</i>	IPSS OBB 1.0.0	<i>Scope:</i>	Sub-System
<i>See also:</i>	[15] IPS:SSS:REQ:8001	<i>Originator:</i>	Technics (R&D)
<i>Verification method:</i>	Test	<i>Verification step:</i>	System test
<i>Issues/Decisions:</i>			

EL4-IPS-SSS-REQ-08009- Interface to adjacent [VIL]s - X25 over RaSTA FUR			
IPS shall provide connectivity to [Adjacent Interlocking] system.			
<i>Comments:</i> The communication protocol is [X25 over RaSTA] or X25 over RaSTA over TLS.			
<i>Legacy ID:</i>		<i>Priority:</i>	Mandatory
<i>Status:</i>	Approved	<i>Origin:</i>	EL4-DIL-SSS-REQ-00472 EL4-DIL-SSS-REQ-00732
<i>Target Release:</i>	IPSS OBB 1.0.0	<i>Scope:</i>	Sub-System
<i>See also:</i>	[15] IPS:SSS:REQ:8001	<i>Originator:</i>	Technics (R&D)
<i>Verification method:</i>	Test	<i>Verification step:</i>	System test
<i>Issues/Decisions:</i>			

3.8 INTERNAL INTERFACES

All internal interface requirements are listed here.

EL4-IPS-SSS-REQ-08101- IF-70 Workflow Scenarios - IPS Cloud FUR			
IPS shall provide interface IF-70 RETROFIT scenarios - IPS Cloud.			
<i>Comments:</i> The interface IF-70 is specified in document [02].			
<i>Legacy ID:</i>	IPS:SSS:REQ:8101	<i>Priority:</i>	Mandatory
<i>Status:</i>	Approved	<i>Origin:</i>	EL4-DIL-SSS-REQ-00477
<i>Target Release:</i>	IPSS OBB 1.0.0 BFZ-3.0	<i>Scope:</i>	Sub-System
<i>See also:</i>		<i>Originator:</i>	Technics (R&D)
<i>Verification method:</i>	Test	<i>Verification step:</i>	System test
<i>Issues/Decisions:</i>			

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EL4-IPS-SSS-REQ-08102- IF-71 Container Delivery Items - IPS Cloud FUR			
IPS shall provide rules via interface IF-71 defining how the containers shall be delivered.			
<i>Comments:</i> The interface IF-71 is specified in document [03].			
<i>Legacy ID:</i>	IPS:SSS:REQ:8102	<i>Priority:</i>	Mandatory
<i>Status:</i>	Approved	<i>Origin:</i>	EL4-FL-FEA-0015
<i>Target Release:</i>	IPSS OBB 1.0.0 BFZ-3.0	<i>Scope:</i>	Sub-System
<i>See also:</i>		<i>Originator:</i>	Technics (R&D)
<i>Verification method:</i>	Test	<i>Verification step:</i>	System test
<i>Issues/Decisions:</i>			

EL4-IPS-SSS-REQ-08103- IF-72 Prometheus (ADM - IPS Cloud) FUR			
IPS shall implement interface IF-72 ADM - IPS Cloud monitoring.			
<i>Comments:</i> The interface IF-72 is specified in document [04].			
<i>Legacy ID:</i>	IPS:SSS:REQ:8103	<i>Priority:</i>	Mandatory
<i>Status:</i>	Approved	<i>Origin:</i>	EL4-DIL-SSS-REQ-00476 EL4-DIL-SSS-REQ-00321 EL4-DIL-SSS-REQ-00990 EL4-DIL-SSS-REQ-00329 EL4-DIL-SSS-REQ-00330
<i>Target Release:</i>	IPSS OBB 1.0.0 BFZ-3.0	<i>Scope:</i>	Sub-System
<i>See also:</i>		<i>Originator:</i>	Technics (R&D)
<i>Verification method:</i>	Test	<i>Verification step:</i>	System test
<i>Issues/Decisions:</i>			

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EL4-IPS-SSS-REQ-08105- IF-73 Grafana (ADM - IPS Cloud) FUR			
IPS shall implement interface IF-73 Grafana (ADM - IPS Cloud).			
<i>Comments:</i> The interface IF-73 is specified in document [09].			
<i>Legacy ID:</i>	IPS:SSS:REQ:8105	<i>Priority:</i>	Mandatory
<i>Status:</i>	Approved	<i>Origin:</i>	EL4-DIL-SSS-REQ-00476
<i>Target Release:</i>	IPSS OBB 1.0.0 BFZ-3.0	<i>Scope:</i>	Sub-System
<i>See also:</i>		<i>Originator:</i>	Technics (R&D)
<i>Verification method:</i>	Test	<i>Verification step:</i>	System test
<i>Issues/Decisions:</i>			

3.9 SAFETY REQUIREMENTS

According to the TAS Platform and the J4S safety concept, all possible influences of IPS to the safety-related operation of DIL are mitigated. Therefore, IPS can be treated as a "gray channel / grauer Kanal " from the safety point of view. Therefore, there are no safety requirements defined. IPS provides functions to enable VIL to fulfil the safety-related application conditions for SIL-4 operation as required by TAS Platform and J4S.

All safety-related application conditions required by TAS Platform and J4S will be listed in the Evidence against SRACs for DIL document [26]. Dedicated IPS requirements are defined below to fulfil relevant SRACs or point(s) within SRACs.

3.9.1 IPS REQUIREMENTS RELATED TO TAS-PLF CORE 2.X HW AND J4S SRACs

EL4-IPS-SSS-REQ-08202- Provision of IIS Container on Compute Nodes FUR			
IPS BASE shall provide an IIS container (container image from J4S) on each compute node which hosts safety-related virtualized applications.			
<i>Comments:</i>			
<i>Legacy ID:</i>	IPS:SSS:REQ:8202	<i>Priority:</i>	Mandatory
<i>Status:</i>	Approved	<i>Origin:</i>	EL4-DIL-SSS-REQ-00437
<i>Target Release:</i>	IPSS OBB 1.0.0 BFZ-3.0	<i>Scope:</i>	Sub-System
<i>See also:</i>		<i>Originator:</i>	Technics (R&D)
<i>Verification method:</i>	Test	<i>Verification step:</i>	Integration test
<i>Issues/Decisions:</i>			

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EL4-IPS-SSS-REQ-08203- Provision of OCC Configuration for each IIS Container FUR			
IPS BASE shall provide for each IIS container the OCC configuration file.			
<i>Comments:</i>			
<i>Legacy ID:</i>	IPS:SSS:REQ:8203	<i>Priority:</i>	Mandatory
<i>Status:</i>	Approved	<i>Origin:</i>	EL4-DIL-SSS-REQ-00437
<i>Target Release:</i>	IPSS OBB 1.0.0 BFZ-3.0	<i>Scope:</i>	Sub-System
<i>See also:</i>		<i>Originator:</i>	Technics (R&D)
<i>Verification method:</i>	Test	<i>Verification step:</i>	Integration test
<i>Issues/Decisions:</i>			

3.9.2 IPS REQUIREMENTS RELATED TO TAS-PLF CORE 2.x SW SRACs

EL4-IPS-SSS-REQ-08304- NTP - time drift CSR			
The time drift of the clocks exposed to NMR application instance redundancies shall not cause a safety reaction in the NMR mechanism.			
<i>Comments:</i> Safety reaction limits are specified in the SSRS NMR - TAS Plf. [43]			
<i>Legacy ID:</i>		<i>Priority:</i>	Mandatory
<i>Status:</i>	Approved	<i>Origin:</i>	EL4-DIL-SSS-REQ-00560 EL4:IDJ-RA:ID:0003
<i>Target Release:</i>	IPSS OBB 1.0.0	<i>Scope:</i>	Sub-System
<i>See also:</i>	[43]PLF-2.x:SSRS:NMR:0048	<i>Originator:</i>	Technics (R&D)
<i>Verification method:</i>	Test	<i>Verification step:</i>	Integration test
<i>Issues/Decisions:</i>	EL4:IDJ-RA:ID:0003 Usage of virtualized TAS Plf. 2.6.x Decision: In the DSTW Concept V04 it is decided to use TAS Plf. 2.6.x as the operating system for Virtual Elektra (VIL) running as Virtual Machines on top of IPS.		

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EL4-IPS-SSS-REQ-08305- Disk image for NMR application instance	
FUR	
IPS shall ensure that all instances of a single NMR application use the same disk image.	
Comments:	
Legacy ID:	Priority: Mandatory
Status: Approved	Origin: EL4:IDJ-RA:ID:0003 EL4-DIL-SSS-REQ-01283
Target Release: IPSS OBB 1.0.0	Scope: Sub-System
See also: [43]PLF-2.x:SSRS:NMR:0094	Originator: Technics (R&D)
Verification method: Test	Verification step: Integration test
Issues/Decisions: EL4:IDJ-RA:ID:0003 Usage of virtualized TAS Plf. 2.6.x Decision: In the DSTW Concept V04 it is decided to use TAS Plf. 2.6.x as the operating system for Virtual Elektra (VIL) running as Virtual Machines on top of IPS.	

EL4-IPS-SSS-REQ-08306- NMR application instances - redundant connection	
DPR	
IPS shall provide redundant communication path between NMR application instances.	
Comments:	
Legacy ID:	Priority: Mandatory
Status: Approved	Origin: EL4:IDJ-RA:ID:0003 EL4-DIL-SSS-REQ-00326 EL4-DIL-SSS-REQ-00328 EL4-DIL-SSS-REQ-01473
Target Release: IPSS OBB 1.0.0	Scope: Sub-System
See also: [43]PLF-2.x:SSRS:NMR:0003 [43]PLF-2.x:SSRS:NMR:0121 EL4-IPS-SSS-REQ-00019	Originator: Technics (R&D)
Verification method: Test	Verification step: Integration test
Issues/Decisions: EL4:IDJ-RA:ID:0003 Usage of virtualized TAS Plf. 2.6.x Decision: In the DSTW Concept V04 it is decided to use TAS Plf. 2.6.x as the operating system for Virtual Elektra (VIL) running as Virtual Machines on top of IPS.	

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EL4-IPS-SSS-REQ-08307- PRR NMR application instances - interconnectivity			
IPS shall provide highly prioritized, low-latency NMR application interconnectivity.			
<i>Comments:</i>			
<i>Legacy ID:</i>		<i>Priority:</i>	Mandatory
<i>Status:</i>	Approved	<i>Origin:</i>	EL4:IDJ-RA:ID:0003
<i>Target Release:</i>	IPSS OBB 1.0.0	<i>Scope:</i>	Sub-System
<i>See also:</i>	[43]PLF-2.x:SSRS:NMR:0113	<i>Originator:</i>	Technics (R&D)
<i>Verification method:</i>	Test	<i>Verification step:</i>	Integration test
<i>Issues/Decisions:</i> EL4:IDJ-RA:ID:0003 Usage of virtualized TAS Plf. 2.6.x Decision: In the DSTW Concept V04 it is decided to use TAS Plf. 2.6.x as the operating system for Virtual Elektra (VIL) running as Virtual Machines on top of IPS.			

EL4-IPS-SSS-REQ-08308- PRR NMR network traffic - size			
IPS shall provide NMR network traffic with a minimum of MTU size of 16kbytes.			
<i>Comments:</i>			
<i>Legacy ID:</i>		<i>Priority:</i>	Mandatory
<i>Status:</i>	Approved	<i>Origin:</i>	EL4:IDJ-RA:ID:0003
<i>Target Release:</i>	IPSS OBB 1.0.0	<i>Scope:</i>	Sub-System
<i>See also:</i>	[43]PLF-2.x:SSRS:NMR:0050	<i>Originator:</i>	Technics (R&D)
<i>Verification method:</i>	Test	<i>Verification step:</i>	Integration test
<i>Issues/Decisions:</i> EL4:IDJ-RA:ID:0003 Usage of virtualized TAS Plf. 2.6.x Decision: In the DSTW Concept V04 it is decided to use TAS Plf. 2.6.x as the operating system for Virtual Elektra (VIL) running as Virtual Machines on top of IPS.			

EL4-IPS-SSS-REQ-08309- PRR NMR network traffic - latency			
IPS shall support georedundant NMR traffic with less than 30ms latency.			
<i>Comments:</i>			
<i>Legacy ID:</i>		<i>Priority:</i>	Mandatory
<i>Status:</i>	Approved	<i>Origin:</i>	EL4:IDJ-RA:ID:0003
<i>Target Release:</i>	IPSS OBB 1.0.0	<i>Scope:</i>	Sub-System
<i>See also:</i>	[43]PLF-2.x:SSRS:NMR:0122	<i>Originator:</i>	Technics (R&D)
<i>Verification method:</i>	Test	<i>Verification step:</i>	Integration test
<i>Issues/Decisions:</i> EL4:IDJ-RA:ID:0003 Usage of virtualized TAS Plf. 2.6.x Decision: In the DSTW Concept V04 it is decided to use TAS Plf. 2.6.x as the operating system for Virtual Elektra (VIL) running as Virtual Machines on top of IPS.			

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EL4-IPS-SSS-REQ-08310- PRR Provision of CDS	
IPS shall provide [CDS] per Datacenter	
<i>Comments:</i>	
<i>Legacy ID:</i> <i>Status:</i> Approved <i>Target Release:</i> IPSS OBB 1.0.0 <i>See also:</i> [43]PLF-2.x:SSRS:NMR:0122 <i>Verification method:</i> Test	<i>Priority:</i> Mandatory <i>Origin:</i> EL4:IDJ-RA:ID:0003 EL4-DIL-SSS-REQ-01283 <i>Scope:</i> Sub-System <i>Originator:</i> Technics (R&D) <i>Verification step:</i> Integration test
<i>Issues/Decisions:</i> EL4:IDJ-RA:ID:0003 Usage of virtualized TAS Plf. 2.6.x Decision: In the DSTW Concept V04 it is decided to use TAS Plf. 2.6.x as the operating system for Virtual Elektra (VIL) running as Virtual Machines on top of IPS.	

3.9.3 INTERFACE - INTEGRITY LEVELS

EL4-IPS-SSS-REQ-08316- SAR Safety - Integrity Levels - Interface [VIL] - [IPS]	
[VIL] - [IPS] interfaces shall meet the requirements for basic integrity [BI]	
<i>Comments:</i>	
<i>Legacy ID:</i> <i>Status:</i> Approved <i>Target Release:</i> IPSS OBB 1.0.0 <i>See also:</i> <i>Verification method:</i> Test	<i>Priority:</i> Mandatory <i>Origin:</i> EL4-DIL-SSS-REQ-00685 <i>Scope:</i> Sub-System <i>Originator:</i> Technics (R&D) <i>Verification step:</i> Integration test
<i>Issues/Decisions:</i>	

3.10 COMPUTER HARDWARE REQUIREMENTS

Other hardware requirements are defined in Generic Server and Network Cabinet SSS [06].

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EL4-IPS-SSS-REQ-08801- Network Cabinet PER			
The infrastructure platform (OpenStack abstractions of networking capabilities) provided by IPS shall be capable of running on the network components in the network cabinet defined in the Generic Server and Network Cabinet SSS [06] and mentioned in SSDD Generic Server and Network Cabinet [12].			
<i>Comments:</i>			
<i>Legacy ID:</i>	IPS:SSS:REQ:8801	<i>Priority:</i>	Mandatory
<i>Status:</i>	Approved	<i>Origin:</i>	EL4-FL-FEA-0015
<i>Target Release:</i>	IPSS OBB 1.0.0 BFZ-3.0	<i>Scope:</i>	Sub-System
<i>See also:</i>		<i>Originator:</i>	Technics (R&D)
<i>Verification method:</i>	Review	<i>Verification step:</i>	Review
<i>Issues/Decisions:</i>			

EL4-IPS-SSS-REQ-08802- Server Cabinet PER			
The infrastructure platform (OpenStack abstractions of computing, storage and networking capabilities) provided by IPS shall be capable of running on the hardware components (compute nodes, storage nodes, control nodes, RAID, internal network components) in the computing cabinets defined in the Generic Server and Network Cabinet SSS [06] and mentioned in SSDD Generic Server and Network Cabinet [12].			
<i>Comments:</i> The network connectivity is covered by requirement EL4-IPS-SSS-REQ-01001.			
<i>Legacy ID:</i>	IPS:SSS:REQ:8802	<i>Priority:</i>	Mandatory
<i>Status:</i>	Approved	<i>Origin:</i>	EL4-DIL-SSS-REQ-00481
<i>Target Release:</i>	IPSS OBB 1.0.0 BFZ-3.0	<i>Scope:</i>	Sub-System
<i>See also:</i>		<i>Originator:</i>	Technics (R&D)
<i>Verification method:</i>	Review	<i>Verification step:</i>	Review
<i>Issues/Decisions:</i>			

3.11 DESIGN AND IMPLEMENTATION CHARACTERISTICS

3.11.1 RAM-DESIGN RULES

All RAM related requirements for IPS are listed here. For the list of used terms within this chapter refer to the top level SSS [01].

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EL4-IPS-SSS-REQ-09902- Server failure DPR			
IPS shall redistribute remaining services following the balanced increment principle in case of a server failure.			
<i>Comments:</i> There are dedicated retrofit scenarios for that in case of failure that service are distributed on remaining other servers. This ensures a balanced distribution of the services on the remaining servers.			
<i>Legacy ID:</i>	IPS:SSS:REQ:9902	<i>Priority:</i>	Mandatory
<i>Status:</i>	Approved	<i>Origin:</i>	EL4-FL-FEA-0015 EL4-DIL-SSS-REQ-00322 EL4-DIL-SSS-REQ-00323 EL4-DIL-SSS-REQ-00328 EL4-DIL-SSS-REQ-01485
<i>Target Release:</i>	IPSS OBB 1.0.0 BFZ-3.0	<i>Scope:</i>	Sub-System
<i>See also:</i>		<i>Originator:</i>	Technics (R&D)
<i>Verification method:</i>	Test	<i>Verification step:</i>	Integration test
<i>Issues/Decisions:</i>			

EL4-IPS-SSS-REQ-09903- Server reintegration DPR			
IPS shall not redistribute services in case of server reintegration to prevent any additional disruption of operating services.			
<i>Comments:</i> This means that services redistributed in case of server failure, shall stay running as is, regardless of server reintegration.			
There are dedicated retrofit scenarios for that in case of failure that services are distributed on remaining other servers.			
<i>Legacy ID:</i>	IPS:SSS:REQ:9903	<i>Priority:</i>	Mandatory
<i>Status:</i>	Approved	<i>Origin:</i>	EL4-FL-FEA-0015 EL4-DIL-SSS-REQ-00322 EL4-DIL-SSS-REQ-00323 EL4-DIL-SSS-REQ-01486
<i>Target Release:</i>	IPSS OBB 1.0.0 BFZ-3.0	<i>Scope:</i>	Sub-System
<i>See also:</i>		<i>Originator:</i>	Technics (R&D)
<i>Verification method:</i>	Test	<i>Verification step:</i>	Integration test
<i>Issues/Decisions:</i>			

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EL4-IPS-SSS-REQ-09906- APP-Layer - substation specific services: colocation/grouping DPR			
IPS shall execute a group of services which belongs to a specific sub-station on the same physical server.			
<i>Comments:</i>			
<i>Legacy ID:</i>	IPS:SSS:REQ:9906	<i>Priority:</i>	Mandatory
<i>Status:</i>	In negotiation	<i>Origin:</i>	EL4-FL-FEA-0015 EL4-DIL-SSS-REQ-01482
<i>Target Release:</i>	IPSS OBB 1.0.0 BFZ-3.0	<i>Scope:</i>	Sub-System
<i>See also:</i>		<i>Originator:</i>	Technics (R&D)
<i>Verification method:</i>	Test	<i>Verification step:</i>	Integration test
<i>Issues/Decisions:</i>			

EL4-IPS-SSS-REQ-09910- APP-Layer - global services: balanced distribution DPR			
IPS shall place global service instances on different physical servers and different availability zones following the balanced distribution principle.			
<i>Comments:</i> Using the following Kubernetes schedule mechanism at each global application pod level (defined within the global application deployment file) ensures that global application pods are distributed across the three availability zones. •Pod anti-affinity: global application pods are not scheduled on the same VM •Topology spread constraints: global application pods are distributed across the three availability zones			
<i>Legacy ID:</i>	IPS:SSS:REQ:9910	<i>Priority:</i>	Mandatory
<i>Status:</i>	Approved	<i>Origin:</i>	EL4-FL-FEA-0015 EL4-DIL-SSS-REQ-00322 EL4-DIL-SSS-REQ-00323 EL4-DIL-SSS-REQ-00328 EL4-DIL-SSS-REQ-01485
<i>Target Release:</i>	IPSS OBB 1.0.0 BFZ-3.0	<i>Scope:</i>	Sub-System
<i>See also:</i>		<i>Originator:</i>	Technics (R&D)
<i>Verification method:</i>	Test	<i>Verification step:</i>	Integration test
<i>Issues/Decisions:</i>			

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EL4-IPS-SSS-REQ-09911- IPS INFRA - balanced distribution DPR			
IPS shall place IPS service instances belonging to the IPS INFRA Cluster on different servers and different availability zones following the balanced distribution principle.			
<i>Comments:</i> All IPS components within the INFRA cluster will be deployed as Statefulset with 3 Replicas each running on a different worker VM on a different availability zone.			
<i>Legacy ID:</i>	IPS:SSS:REQ:9911	<i>Priority:</i>	Mandatory
<i>Status:</i>	Approved	<i>Origin:</i>	EL4-FL-FEA-0015 EL4-DIL-SSS-REQ-00322 EL4-DIL-SSS-REQ-00323 EL4-DIL-SSS-REQ-00328 EL4-DIL-SSS-REQ-01485
<i>Target Release:</i>	IPSS OBB 1.0.0 BFZ-3.0	<i>Scope:</i>	Sub-System
<i>See also:</i>		<i>Originator:</i>	Technics (R&D)
<i>Verification method:</i>	Test	<i>Verification step:</i>	Integration test
<i>Issues/Decisions:</i>			

EL4-IPS-SSS-REQ-09912- IPS INFRA - high availability DPR			
IPS shall execute IPS service instances belonging to the IPS INFRA Cluster 2003 hot-redundant or 1002 hot-redundant or 1003 hot-redundant.			
<i>Comments:</i> All IPS components within the INFRA cluster will be deployed as Statefulset with 3 Replicas each running on a different worker VM on a different availability zone.			
<i>Legacy ID:</i>	IPS:SSS:REQ:9912	<i>Priority:</i>	Mandatory
<i>Status:</i>	Approved	<i>Origin:</i>	EL4-FL-FEA-0015 EL4-DIL-SSS-REQ-00322 EL4-DIL-SSS-REQ-00323 EL4-DIL-SSS-REQ-00328 EL4-DIL-SSS-REQ-00406 EL4-DIL-SSS-REQ-00407 EL4-DIL-SSS-REQ-00408
<i>Target Release:</i>	IPSS OBB 1.0.0 BFZ-3.0	<i>Scope:</i>	Sub-System
<i>See also:</i>		<i>Originator:</i>	Technics (R&D)
<i>Verification method:</i>	Test	<i>Verification step:</i>	Integration test
<i>Issues/Decisions:</i>			

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EL4-IPS-SSS-REQ-09913- IPS SERVICE - balanced distribution DPR			
IPS shall place IPS Service instances belonging to an IPS SERVICE Cluster on different servers and different availability zones following the balanced distribution principle.			
<i>Comments:</i> All IPS components within the SERVICE cluster will be deployed as Statefulset with 3 Replicas each running on a different worker VM on a different availability zone.			
<i>Legacy ID:</i>	IPS:SSS:REQ:9913	<i>Priority:</i>	Mandatory
<i>Status:</i>	Approved	<i>Origin:</i>	EL4-FL-FEA-0015 EL4-DIL-SSS-REQ-00322 EL4-DIL-SSS-REQ-00323 EL4-DIL-SSS-REQ-00328 EL4-DIL-SSS-REQ-01485
<i>Target Release:</i>	IPSS OBB 1.0.0 BFZ-3.0	<i>Scope:</i>	Sub-System
<i>See also:</i>		<i>Originator:</i>	Technics (R&D)
<i>Verification method:</i>	Test	<i>Verification step:</i>	Integration test
<i>Issues/Decisions:</i>			

EL4-IPS-SSS-REQ-09914- IPS SERVICE - high availability DPR			
IPS shall execute IPS Service instances belonging to an IPS SERVICE Cluster 2oo3 hot-redundant or 1oo2 hot-redundant or 1oo3 hot-redundant.			
<i>Comments:</i> All IPS components within the SERVICE cluster will be deployed as Statefulset with 3 Replicas each running on a different worker VM on a different availability zone.			
<i>Legacy ID:</i>	IPS:SSS:REQ:9914	<i>Priority:</i>	Mandatory
<i>Status:</i>	Approved	<i>Origin:</i>	EL4-FL-FEA-0015 EL4-DIL-SSS-REQ-00322 EL4-DIL-SSS-REQ-00323 EL4-DIL-SSS-REQ-00328 EL4-DIL-SSS-REQ-00406 EL4-DIL-SSS-REQ-00407 EL4-DIL-SSS-REQ-00408
<i>Target Release:</i>	IPSS OBB 1.0.0 BFZ-3.0	<i>Scope:</i>	Sub-System
<i>See also:</i>		<i>Originator:</i>	Technics (R&D)
<i>Verification method:</i>	Test	<i>Verification step:</i>	Integration test
<i>Issues/Decisions:</i>			

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EL4-IPS-SSS-REQ-09915- IPS BASE - CGs balanced distribution DPR			
IPS shall place CGs on different physical servers and on different availability zones following the balanced distribution principle.			
<i>Comments:</i>			
<i>Legacy ID:</i>		<i>Priority:</i>	Mandatory
<i>Status:</i>	Approved	<i>Origin:</i>	EL4-DIL-SSS-REQ-01490
<i>Target Release:</i>	IPSS OBB 1.0.0	<i>Scope:</i>	Sub-System
<i>See also:</i>		<i>Originator:</i>	Technics (R&D)
<i>Verification method:</i>	Test	<i>Verification step:</i>	Integration test
<i>Issues/Decisions:</i>			

EL4-IPS-SSS-REQ-09916- IPS BASE - CNs balanced distribution DPR			
IPS shall place CNs belonging to the same CG on different physical servers and on different availability zones following the balanced distribution principle.			
<i>Comments:</i>			
<i>Legacy ID:</i>		<i>Priority:</i>	Mandatory
<i>Status:</i>	Approved	<i>Origin:</i>	EL4-DIL-SSS-REQ-01491
<i>Target Release:</i>	IPSS OBB 1.0.0	<i>Scope:</i>	Sub-System
<i>See also:</i>		<i>Originator:</i>	Technics (R&D)
<i>Verification method:</i>	Test	<i>Verification step:</i>	Integration test
<i>Issues/Decisions:</i>			

EL4-IPS-SSS-REQ-09917- IPS BASE - CEs balanced distribution DPR			
IPS shall place CEs belonging to a CN on different physical servers and on different availability zones following the balanced distribution principle.			
<i>Comments:</i>			
<i>Legacy ID:</i>		<i>Priority:</i>	Mandatory
<i>Status:</i>	In negotiation	<i>Origin:</i>	EL4-DIL-SSS-REQ-01492
<i>Target Release:</i>	IPSS OBB 1.0.0	<i>Scope:</i>	Sub-System
<i>See also:</i>		<i>Originator:</i>	Technics (R&D)
<i>Verification method:</i>	Test	<i>Verification step:</i>	Integration test
<i>Issues/Decisions:</i>			

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EL4-IPS-SSS-REQ-09918- DPR Server cabinet failure	
[IPS] shall support uninterrupted operation in the event of a complete failure of the server cabinet (resp. a complete availability zone).	
Comments:	
Legacy ID:	Priority: Mandatory
Status: Approved	Origin:
Target Release: IPSS OBB 1.0.0	Scope: Sub-System
See also:	Originator: Technics (R&D)
Verification method: Test	Verification step: System test
Issues/Decisions:	

3.12 DELIVERY RELATED REQUIREMENTS

EL4-IPS-SSS-REQ-09102- DOR Documentation and user manuals	
[IPS] shall provide end-user related support documentation and user manuals in the printed and electronic form.	
Comments:	
Legacy ID:	Priority: Mandatory
Status: Approved	Origin: EL4-DIL-SSS-REQ-00345 EL4-DIL-SSS-REQ-00346 EL4-DIL-SSS-REQ-00347 EL4-DIL-SSS-REQ-00348
Target Release: IPSS OBB 1.0.0	Scope: Sub-System
See also:	Originator: Technics (R&D)
Verification method: Review	Verification step: Review
Issues/Decisions:	

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EL4-IPS-SSS-REQ-09701-Geo-redundancy - Roll-out - Tools and Supporting Documentation DOR			
Roll-out to the customer shall be supported by tools, installation guidelines and user manuals.			
<i>Comments:</i> The Installation Guide IPS is defined in [32]. The rollout procedure is defined in Rollout Concept [40] and in chapter 6.5 of SSDD IPS [28] document.			
<i>Legacy ID:</i>	IPS:SSS:REQ:9701	<i>Priority:</i>	Mandatory
<i>Status:</i>	Approved	<i>Origin:</i>	EL4-DIL-SSS-REQ-00346
<i>Target Release:</i>	IPSS OBB 1.0.0 BFZ-3.0	<i>Scope:</i>	Sub-System
<i>See also:</i>		<i>Originator:</i>	Technics (R&D)
<i>Verification method:</i>	Review	<i>Verification step:</i>	Review
<i>Issues/Decisions:</i>			

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3.13 OPEN ISSUES

1. The references in Chapter 1.6 Referenced Documents which point to documents from the [OMC] product need further analysis, review and rework, or full replacement by a new document if they are not valid for [DIL].
2. There are missing Values for the "Origin" attribute. This indicates that the link between subsystem and system requirement or feature has not been identified or cannot be established. This gap will be addressed and improved during next FEAs refinement.
3. Generic terminology for REQs reused from 3BU 51500 3200 DTAPA Ed.03 needs to be defined and agreed upon in order to make the infrastructure platform created by the BFZ geo-redundancy concept reusable for other products/security technology systems (e.g. ELEKTRA 4.0, SCWS 2.0, ...). Terms may potentially include but not limited to:
 - "Core Network" - instead of the "OMC 3.0 Core Network "
 - "Datacentre"/ "Location" - instead of "OMC Location"
 - "Instance" - instead of "OMC instance"
 - "Applications " - instead of "OMC applications"
 - "Interconnection Network" - instead of "OMC 3.x Interconnection Network"
 - "Interfaces" - instead of "OMC interfaces"
4. Missing Requirement for restoring service. The current Specification defines configuration requirement for the backup service [EL4-IPS-SSS-REQ-00012], however there is no corresponding requirement covering the restoring service. The restore functionality - the process of recovering data or configurations from backups must also be specified to ensure full data recovery capability.
5. Missing definitions and configurations details for the geo-redundant connections [EL4-IPS-SSS-REQ-00021], [EL4-IPS-SSS-REQ-00022]. They shall be specified in a future requirement increment to ensure proper implementation of the automatic switch -over.
6. Missing definitions which data, configurations, VMs, and images shall be synchronized between datacenters including synchronization frequency, direction, method, validation criteria. [EL4-IPS-SSS-REQ-00024]. These aspects shall be specified in a future requirement increment to ensure proper implementation of the synchronization mechanism.
7. Missing definitions of the list of projects and applications whose performance and capacity requirement IPS shall support. [EL4-IPS-SSS-REQ-00033]. The list shall be defined in a future requirement increment.

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8. The requirements with the status “In negotiation” could not be fully addressed within the scope of this releasedue to pending clarifications and open dependencies (e.g. interface definitions, safety inputs, system-level clarifications, or stakeholders decisions). These requirements will be refined in and incorporated into the next release once the necessary information becomes available,

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4. HISTORY

4.1 EDITION 01

First edition SRR-1 milestone prior to contract signature.

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APPENDIX A DECISIONS

This chapter shall contain important decisions made in connection with the specified requirements.

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